

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree Examination December 2022 (2019 scheme)

Course Code: EET401**Course Name: ADVANCED CONTROL SYSTEMS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- 1 What do you mean by (i) state variables (ii) phase variables and (iii) state space? (3)
- 2 A series RLC circuit is excited by a voltage source, $v(t)$ volts and the output is measured across the resistor. Derive the state model of the electrical system. (3)
- 3 State and prove any three properties of state transition matrix. (3)
- 4 The state space representation and transfer function of a system are given below. Find the value of 'm'. (3)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\frac{Y(s)}{U(s)} = \frac{2s+2}{(s+1)(s+4)}$$
- 5 With necessary equations, explain duality principle. (3)
- 6 Explain the Gilbert's method to test controllability of n^{th} order system represented by the state model given below. Assume that the eigen values are distinct. (3)

$$\dot{X} = AX + BU \quad Y = CX + DU$$
- 7 Write any three characteristics of non-linear systems. (3)
- 8 Explain the following (i) jump resonance (ii) stable limit cycle and (iii) unstable limit cycle? (3)
- 9 Define (i) singular point (ii) vortex point and (iii) saddle point. (3)
- 10 State the condition/s for a scalar function $V(x)$ to be (i) positive definite (ii) negative definite and (iii) indefinite. Give one example for each. (3)

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

- 11 a) From basics, derive the state model of an armature controlled dc motor. (7)
- b) Differential equation of the system is given by, (7)

$$\frac{d^3 y}{dt^3} + 6 \frac{d^2 y}{dt^2} + 11 \frac{dy}{dt} + 6y + u = 0$$

where, y is the output and u is the input. Derive the state model of the system using phase variables.

OR

- 12 a) What is similarity transformation? How similarity transformation is helpful for state space analysis? (5)
- b) Obtain the diagonal canonical form of the state model given below. (9)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} u \quad [y] = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Module II

- 13 A discrete time system is given by, (14)
- $$y(k+2) + 5y(k+1) + 6y(k) = u(k)$$

where, y is the output and u is the input. Assume zero initial conditions and the sampling period as 1 second. (i) Derive the state model of the system in Jordan canonical form (ii) Draw the block diagram of the state model and (iii) Compute the state transition matrix.

OR

- 14 a) State equation of a system is given below. Compute the state transition matrix using Laplace transform method. Also, compute the solution of the given state equation if the initial state vector is $X(0)$. (7)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

- b) Find the state transition matrix using Cayley-Hamilton theorem for the system matrix given below. (7)

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

Module III

- 15 a) Using Kalman's method, check the observability of the system given below. (6)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad [y] = [40 \quad 10 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- b) Design the observer feed back gain matrix for the desired eigen values at $-4, -3 \pm j1$ for the state model of the system given below. (8)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 1 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

OR

- 16 a) Using PBH test, check the controllability of the system given below. (6)

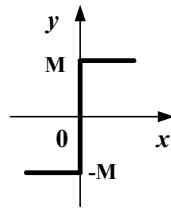
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

- b) Design a feed back controller with state feed back so that the closed loop poles are at -2, -1±j1. The state equation of the original systems is given below. (8)

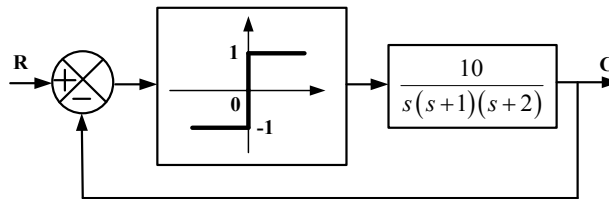
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u$$

Module IV

- 17 a) Derive the describing function of ideal relay non-linearity given below. (6)

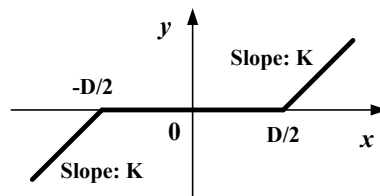


- b) Determine the frequency and nature of the limit cycle for the unity feed back system given below. (8)

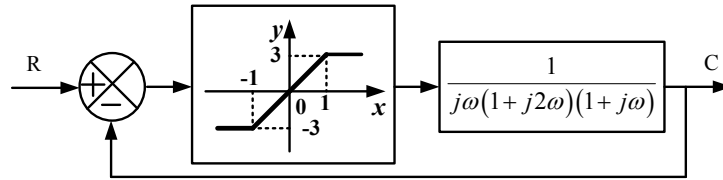


OR

- 18 a) Derive the describing function of dead-zone non-linearity given below. (6)



- b) Determine the frequency and nature of the limit cycle for the unity feed back system given below. (8)

**Module V**

- 19 a) What do you mean by phase trajectory? Explain how to draw the phase trajectory using isocline method. (7)
- b) Compute the Lyapunov function, $V(X)$ for which the system given below is asymptotically stable. (7)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

OR

- 20 a) Construct the phase trajectory of the second order system given below using isocline method. (7)

$$\dot{x}_2 + (2\xi\omega_n)x_2 + (\omega_n^2)x_1 = 0$$

$$X(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1.5 \\ 0 \end{bmatrix} \quad \xi = 0.15 \quad \omega_n = 1 \text{ rad/s}$$

where x_1 and x_2 are the state variables.

- b) Explain Lyapunov function? Using Lyapunov function, $V(X)$, state the condition/s for the system to be (i) stable (ii) asymptotically stable in the large and (iii) unstable. (7)

Consider the system described by,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 - x_2^3$$

Check the stability of the system for the Lyapunov function, $V(X) = x_1^2 + x_2^2$.

Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
B.Tech Degree S7 (R, S) / S7 (PT) (R) Examination December 2023 (2019 Scheme)			
Course Code: EET401			
Course Name: ADVANCED CONTROL SYSTEMS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions, each carries 3 marks.	Marks
1		Definition of phase variable – 1 mark Advantages – 2 marks	(3)
2		Choice of state variables – 1 mark $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} [u]$ – 1 mark $y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ – 1 mark	(3)
3		Transfer function = $C(sI-A)^{-1}B + D$ – 1 mark Calculation – 1 mark Final answer – $2/(s+2)$ – 1 mark	(3)
4		$X(t) = e^{At}X_0$ – 1 mark Finding e^{At} – 1 mark $X(t) = e^{At} X_0 = \begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} e^t \\ te^t \end{bmatrix}$ – 1 mark	(3)

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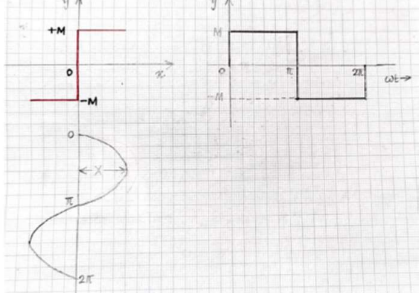
5		Definition of controllability – 1 mark Kalman's method with equations – 2 marks	(3)
6		Explanation of PBH test for observability – 1marks. Equations – 2 marks	(3)
7		Any three non-linearities with figure – 1 mark each	(3)
8		Definition of limit cycle – 1 mark Stable limit cycle with figure – 1mark Unstable limit cycle with figure – 1mark	(3)
9		Definition of singular points – 1 mark Types of singular points with explanation – 2 marks.	(3)
10		(i) positive definite & example (1 mark) (ii) negative definite & example (1 mark) and (iii) indefinite & example (1 mark)	(3)
PART B			
<i>Answer any one full question from each module, each carries 14 marks.</i>			
Module I			
11	a)	KVL equation - 1 mark Torque equation – 1 mark Eqn. for electro mechanical system – 1 mark Select suitable state variables and output variables, substitute them in the above three equations, rearrange the terms. (2 marks) Final state equation and Final output equation (1 mark) Block diagram – 1 mark	(7)
	b)	Rearranging the circuit – 1 mark	

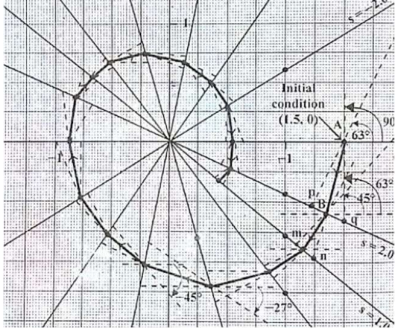
		Node equations and solving for derivative of state variables – 4 marks	(7)
		$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{RC} & \frac{1}{RC} \\ \frac{1}{RC} & -\frac{2}{RC} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{RC} \end{bmatrix} u$ - 1 mark $y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ - 1 mark	
		OR	
12	a)	Characteristic equation – 1 mark $\lambda^3 + 6\lambda^2 + 11\lambda + 6 = 0$ The eigenvalues are $\lambda = -1, -2, -3$ – 1 mark Modal matrix – 3 marks $M = [m_1 \ m_2 \ m_3] = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$ Finding $\Lambda, \tilde{B}, \tilde{C}$ – 3 marks $\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} u \quad ; \quad Y = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$ - 2 marks	(10)
	b)	Any four advantages – 1 mark each	(4)
		Module II	
13	a)	Partial fraction expansion and finding residues – 1 marks Block diagram – 1 mark	(7)

		Finding state equations and output equation – 4 marks Final answer – 1 mark $\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u$ $y = \begin{bmatrix} 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 4u$	
	b)	Characteristic equation – 1 mark Eigen values – 1 mark $f(\lambda_i)$ – 1 mark $f(A)$ – 2 marks $A^k = \begin{bmatrix} 2(-1)^k - (-2)^k & (-1)^k - (-2)^k \\ -2(-1)^k + 2(-2)^k & -(-1)^k + 2(-2)^k \end{bmatrix}$ - 2 marks.	(7)
		OR	
14	a)	$X(t) = e^{At}X_0$ - 1 mark $e^{-At} X(t) = e^{-At} e^{At} X(0)$ $\therefore e^{-At} X(t) = X(0)$ - 1 mark Let $e^{-At} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$ $\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \begin{bmatrix} e^{-2t} \\ -2e^{-2t} \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ - 1 mark	(10)

		$\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ - 1 mark $\therefore e^{-At} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \begin{bmatrix} 2e^t - e^{2t} & e^t - e^{2t} \\ -2e^t + 2e^{2t} & -e^t + 2e^{2t} \end{bmatrix}$ $\therefore e^{At} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$ - 2 marks $\mathbf{L}[e^{At}] = \phi(s)$ Where $\phi(s) = (sI - A)^{-1}$; $\therefore \phi(s)^{-1} = (sI - A)$ or $A = sI - \phi(s)^{-1}$ $\phi(s) = \mathbf{L}[e^{At}] = \begin{bmatrix} \frac{2}{s+1} - \frac{1}{s+2} & \frac{1}{s+1} - \frac{1}{s+2} \\ \frac{-2}{s+1} + \frac{2}{s+2} & \frac{-1}{s+1} + \frac{2}{s+2} \end{bmatrix}$ - 1 mark $\phi(s)^{-1} = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$ - 2 marks $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ - 1 mark	
	b)	Proof of invariance of trace of a matrix	(4)
		Module III	
15	a)		(4)

		Explanation of state observers.	
	b)	Sate equation $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u$ - 3 marks $Q_c = [B \ AB \ A^2B] = \begin{bmatrix} 0 & 0 & 10 \\ 0 & 10 & -30 \\ 10 & -30 & 70 \end{bmatrix}$ - 2 marks Controllability checking - 1 mark Desired characteristic polynomial - 1 mark Determination of gain matrix $K = [0.4 \ 0.4 \ 1]$ - 3 marks	(10)
		OR	
16	a)	$Q_o = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ - 2 marks; Determinant of $Q_o = 1$ - 1 mark Characteristic polynomial of original system, $\lambda^2 + 3\lambda + 1 = 0$ - 2 marks Desired characteristic polynomial, $\lambda^2 + 10\lambda + 25 = 0$ - 1 mark Observer gain matrix, $G = \begin{bmatrix} 7 \\ 10 \end{bmatrix}$ - 4 marks	(10)
	b)	Explanation of duality principle	(4)
		Module IV	

17	a)	 Input signal: $x = X \sin \omega t$ Output signal: $y(\omega t) = \begin{cases} +M & ; 0 < \omega t < \pi \\ -M & ; \pi < \omega t < 2\pi \end{cases}$ Describing function : $A_1 = \frac{4M}{\pi} \quad B_1 = 0 \quad \phi_1 = 0^\circ$ Derivation of constants with necessary steps $K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \quad Y_1 = \sqrt{A_1^2 + B_1^2} \quad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right)$ $K_N(X, \omega) = \frac{4M}{\pi X} \angle 0^\circ$	(6)
	b)	Definition of Describing function – 2 marks Explanation of describing function analysis, concept of enclosure – 6 marks	(8)
		OR	
18	a)	Polar plot with $K = 1$ - 3 marks $-1/K_N$ locus - 2 marks Maximum value of K for the system to be stable ($K=2.25$) - 2 marks The frequency, $\omega = 1/\sqrt{2}$, amplitude = 1 and semi stable limit cycle - 3 marks	(10)
	b)	Any four properties of nonlinear systems.	(4)
		Module V	
19	a)	Determination of singular point – 3 marks Plot of phase trajectory (with necessary steps) - 6 marks Stable focus - 1 mark	(10)

			
	b)	Stable limit cycles - 2 marks Unstable limit cycles - 2 marks	(4)
		OR	
20	a)	Definition of each term - 1 mark each	(4)
	b)	$A^T P + P A = -I$ $\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} + \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$ - 2 marks Solving for P $P = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}$ - 4 marks Positive definiteness of P for stability. $P_1 = \left \frac{3}{2} \right > 0 ; P_2 = \begin{vmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{vmatrix}$ $= \frac{3}{2} - \frac{1}{4} = \frac{6}{4} - \frac{1}{4} = \frac{5}{4} > 0$ - 2 marks Lyapunov function V(x) $V(x) = x^T P x$ $V(x) = (3/2)x_1^2 + x_1 x_2 + x_2^2$ - 2 marks	(10)

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
 Seventh Semester B.Tech Degree (S, FE) Examination May 2023 (2019 Scheme)

Course Code: EET401

Course Name: ADVANCED CONTROL SYSTEMS

Max. Marks: 100

Duration: 3 Hours

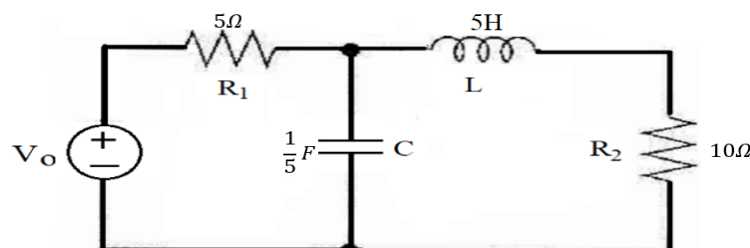
PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
|----|--|-----|
| 1 | Summarize the advantages of state space method over transfer function approach. | (3) |
| 2 | What are phase variables? What are the advantages of choosing phase variables for state space modelling? | (3) |
| 3 | Derive a relation between state model and transfer function of LTI system. | (3) |
| 4 | Write the properties of state transition matrix of discrete time system. | (3) |
| 5 | Define controllability. Explain how can we check the controllability of a system using Kalman's test. | (3) |
| 6 | Illustrate the concept of duality referred to controllability and observability. | (3) |
| 7 | Describe the phenomenon of frequency entrainment in nonlinear systems. | (3) |
| 8 | Derive the describing function of ideal relay nonlinearity. | (3) |
| 9 | Explain what is phase trajectory and phase portrait. | (3) |
| 10 | Determine whether the given quadratic form is positive definite.
$x_1^2 + 3x_2^2 + 11x_3^2 - 2x_1x_2 + 4x_2x_3 + 2x_1x_3$ | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

- 11 a) Obtain the state equation for the network shown in Fig. (7)



- b) Derive the state model of field-controlled dc servo motor. (7)

OR

- 12 a) Develop the state space representation of the system with differential equation (7)

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 10y = 8u(t).$$

- b) Determine the diagonal canonical representation of the system with transfer function (7)

$$T(S) = \frac{2(s+5)}{(s+2)(s+3)(s+4)}$$

Module II

- 13 a) A system is described by following state equation. Find the solution of the state (7)

$$\text{equation. } \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

- b) A discrete time system has the transfer function $\frac{Y(z)}{U(z)} = \frac{4z^3 - 12z^2 + 13z - 7}{(z-1)^2(z-2)}$ (7)

Determine the state model of the system in phase variable form.

OR

- 14 a) Find the state transition matrix using Cayley-Hamilton theorem for the system (8)

$$\text{matrix given below } A = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}$$

- b) Write the transfer function of the system whose state model is given by (6)

$$\dot{x} = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad y = [1 \quad 0] x$$

Module III

- 15 a) Check whether the given system is observable using Gilbert's test. (10)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} u$$

$$y = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- b) What you mean by full order and reduced order observer? Explain. (4)

OR

- 16 a) Evaluate the controllability and observability of the state model using PBH test. (8)

$$\dot{x} = \begin{bmatrix} -5 & 1 \\ -8 & 1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [2 \quad 0] x$$

- b) Illustrate the pole placement technique used for control system design. (6)

Module IV

- 17 a) Define Describing function. Explain how describing function can be used for stability analysis of nonlinear systems. (6)
b) Develop the describing function of relay with dead zone nonlinearity. (8)

OR

- 18 a) Explain any three types of non-linearities that occur in electrical systems. (6)
b) Derive the describing function of saturation nonlinearity. (8)

Module V

- 19 a) What are singular points? Write about the classification of singular points. (6)
b) Describe the Lyapunov's Stability criterion and investigate the stability of the following non-linear systems using Lyapunov's method. (8)

(a) $\dot{x}_1 = -3x_1 + x_2$,

(b) $\dot{x}_2 = -x_1 - x_2 - x_2^3$

OR

- 20 a) Determine the stability of the system described by (10)

$$\dot{x} = Ax, \text{ where } A = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix}$$

Solve for matrix P in the equation $A^T P + P A = -Q$, assuming the matrix Q to be identity matrix

- b) Define the terms (i) stability (ii) asymptotic stability (iii) asymptotically stable in the large (iv) instability. (4)

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B.Tech Degree 7th semester (S,FE) Exam April 2025 (2019 Scheme)

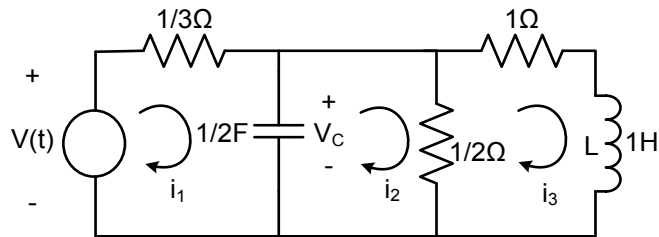
Course Code: EET401**Course Name: ADVANCED CONTROL SYSTEMS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- | | | |
|----|--|-----|
| 1 | How do you choose the state variables for state space modelling? | (3) |
| 2 | Obtain the state model of the system described by the transfer function given as | (3) |
| | $\frac{Y(s)}{U(s)} = \frac{10(S + 4)}{S(S + 1)(S + 3)}$ | |
| 3 | State and prove any three properties of state transition matrix. | (3) |
| 4 | Derive the solution of homogeneous state equation | (3) |
| 5 | Consider the system described by the state model, $\dot{x} = Ax + Bu$ and $y = Cx$ | (3) |
| | A state feedback controller is employed with control law $u = -kx$ where k is the state feedback controller gain matrix. Obtain the modified state model of the system and give the block diagram. | |
| 6 | Discuss the procedure to test the controllability and observability using the PBH method. | (3) |
| 7 | Give the stability conditions based on describing function analysis. | (3) |
| 8 | How the non-linearities are classified? Give examples | (3) |
| 9 | Explain the terms phase plane, phase trajectory and phase portrait | (3) |
| 10 | Check the sign definiteness of the matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 0 \end{bmatrix}$ | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

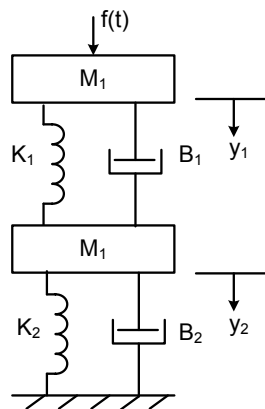
- | | | |
|----|---|-----|
| 11 | a) Derive the state model of the electrical circuit shown below | (9) |
|----|---|-----|



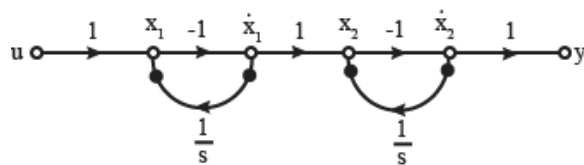
- b) Using the similarity transformation, diagonalize the matrix $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ (5)

OR

- 12 a) Derive the state model of the mechanical system shown below (9)



- b) Determine the state model of the system shown in the signal flow graph given below (5)



Module II

- 13 a) Obtain the unit step response of the system describe by the state model given below for the initial condition, $x(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ (8)

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \ 0] x$$

- b) Compute the state transition matrix using Cayley-Hamilton theorem for the system with system matrix $A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$ (6)

OR

- 14 a) Determine the transfer function of the system describe by the state model given below (8)

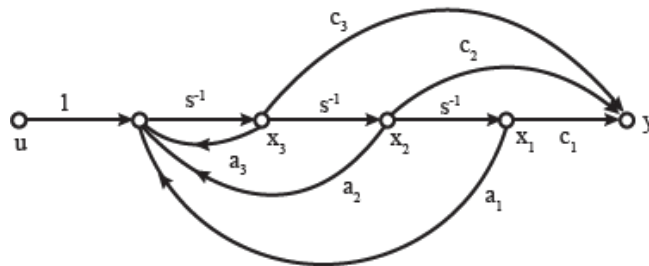
$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$

$$y = [0 \quad 0 \quad 1]x$$

- b) Explain Cayley-Hamilton theorem with an example (6)

Module III

- 15 a) Consider the state space system expressed by the signal flow diagram shown in the figure. Test the controllability using Kalman's method (7)



- b) A regulator has the plant described in the state model given below (7)

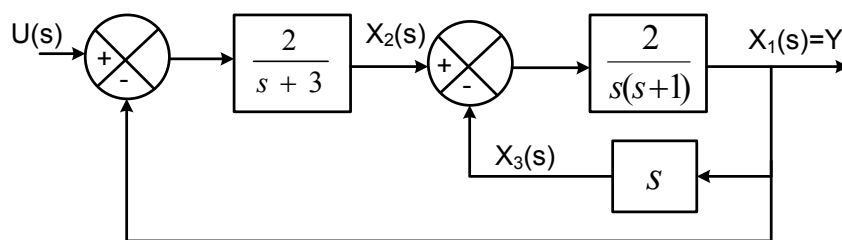
$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 20.6 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 0]x$$

Design a state feedback controller with the control law $u = -KX$ such that the closed loop system has the eigenvalues at $-1.8 \pm j2.4$

OR

- 16 a) Test the controllability and observability of the system shown in the block diagram given below (8)



- b) Design a full order state observer for the system described in the state model given below (6)

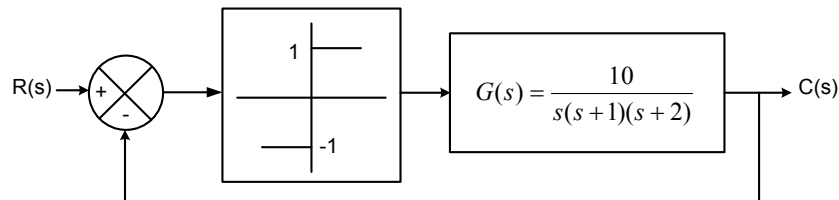
$$\dot{x} = \begin{bmatrix} 0 & 20.6 \\ 1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [0 \quad 1]x$$

The desired eigenvalues of the observer matrix are $\mu_1 = -10$ and $\mu_2 = -10$

Module IV

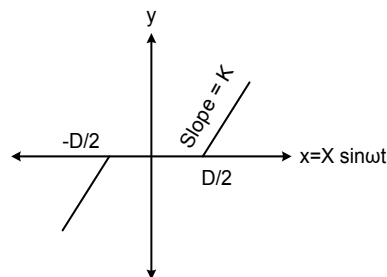
- 17 a) Determine the frequency and nature of the limit cycle for the unity feedback system given below. (10)



- b) State the assumptions made in the describing function method of analysis (4)

OR

- 18 a) Derive the describing function of the non-linearity shown below (8)



- b) Explain the following i) Jump resonance ii) Limit cycle iii) Frequency entrainment (6)

Module V

- 19 a) Explain the stable and unstable limit cycles using necessary sketches. (4)

- b) A linear system is describe by the state equation $\dot{x} = Ax$ where $A = \begin{bmatrix} -4K & 4K \\ 2K & -6K \end{bmatrix}$ (10)

Using the Lyapunov method, find the restrictions on the parameter K to guarantee the system stability.

OR

- 20 a) Determine the singular points of the system described by the differential equation (6)

$$\ddot{x} + 4\dot{x} - 2.5x^2 + 5x = 0$$

- b) Determine the stability of the system described by the state equation given below (8)
using Lyapunov method. Select the Lyapunov function as

$$V(x) = x_1^4 + (x_1 + x_2)^2$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_2 - x_1^3$$

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S7 (R, S) / S7 (PT) (R) Examination December 2023 (2019 Scheme)

Course Code: EET401**Course Name: ADVANCED CONTROL SYSTEMS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- 1 Define phase variables? List the advantages of choosing phase variables for state space modelling. (3)
- 2 Construct the state model for a system characterized by the differential equation (3)

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y + u = 0.$$
- 3 Determine the transfer function of the system with state model (3)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u ; \quad y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$
- 4 A linear time-invariant system is characterized by homogeneous state equation (3)

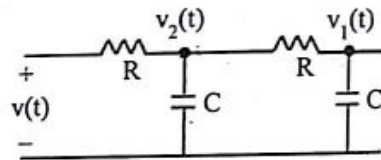
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$
 Compute the solution of the homogeneous equation, assuming the initial state vector, $X_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.
- 5 Define controllability and explain Kalman's method for determining controllability. (3)
- 6 Explain the significance of PBH test for observability. (3)
- 7 Explain the different non linearities with diagram. (3)
- 8 What is limit cycle? How will you determine stable and unstable limit cycle using phase portrait? (3)
- 9 Define singular points? Explain the types of singular points. (3)
 State the condition/s for a scalar function $V(x)$ to be (i) positive definite (ii) negative definite and (iii) indefinite. Give one example for each. (3)
- 10

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

- 11 a) From the basics, derive the state model of a Field controlled dc servo motor and draw the block diagram. (7)
- b) Develop the state model of the electrical network shown in figure by choosing $V_1(t)$ and $V_2(t)$ as state variables. (7)

**OR**

- 12 a) A linear time invariant system is described by the following state model

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} [u] \text{ and } y = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$
 Transform this state model into a canonical state model. (10)
- b) Summarize the advantages of state space method over transfer function approach. (4)

Module II

- 13 a) A discrete time system has the transfer function

$$\frac{Y(Z)}{U(Z)} = \frac{4Z^3 - 12Z^2 + 13Z - 7}{(Z - 1)^2(Z - 2)}$$
 (7)
- Determine the state model of the system in Jordan Canonical form.
- b) The system matrix A of a discrete system is given by $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$. Compute the state transition matrix A^k using Cayley-Hamilton theorem. (7)

OR

- 14 a) For a system represented by state equation
 $\dot{X}(t) = AX(t)$, The response is $X(t) = \begin{bmatrix} e^{-2t} \\ -2e^{-2t} \end{bmatrix}$ when $X(0) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $X(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$ when $X(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Determine the system matrix A and the state transition matrix. (10)
- b) Show that eigen values of state models are unique. (4)

Module III

- 15 a) Describe about state observers. (4)
 b) Consider a linear system described by the transfer function (10)

$$\frac{Y(s)}{U(s)} = \frac{10}{s(s+1)(s+2)}$$

Design a feedback controller with a state feedback so that the closed loop poles are placed at -2, (-1+j1) and (-1-j1).

OR

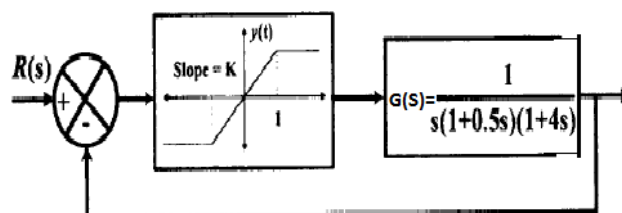
- 16 a) Consider the system described by the state model $\dot{X} = AX$, $y = CX$ where $A = \begin{bmatrix} -1 & 1 \\ 1 & -2 \end{bmatrix}$; $C = [1 \quad 0]$. Design a full-order state observer. The desired eigen values for the observer matrix are $\mu_1 = -5$; $\mu_2 = -5$; (10)
 b) Explain the concept of duality referred to controllability. (4)

Module IV

- 17 a) Develop the describing function of relay nonlinearity. (6)
 b) Define Describing function. Explain how describing function can be used for stability analysis of nonlinear systems. (8)

OR

- 18 a) Determine the value of K for an occurrence of limit cycle. Also determine the frequency, amplitude and stability of limit cycle. (10)



- b) List any four properties of non linear systems. (4)

Module V

- 19 a) A linear second order system is described by the equation (10)
 $\ddot{e} + 2\delta\omega_n\dot{e} + \omega_n^2 e = 0$, Where $\delta = 0.15$, $\omega_n = 1\text{rad/sec}$, $e(0)=1.5$, and $\dot{e}(0) = 0$.
 Determine the singular point and state the stability by constructing the phase trajectory using the method of isoclines.

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- b) Differentiate between stable and unstable limit cycles. (4)

OR

- 20 a) Define the terms (i) stability (ii) asymptotic stability (iii) asymptotically stable in the large (iv) instability (4)

- b) A second order system is represented by $\dot{X} = AX$ where $A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$. (10)

Assuming matrix Q to be identity matrix, solve for matrix P in the equation $A^T P + PA = -Q$. Use Lyapunov theorem and determine the stability of the system. Write the Lyapunov function $V(x)$

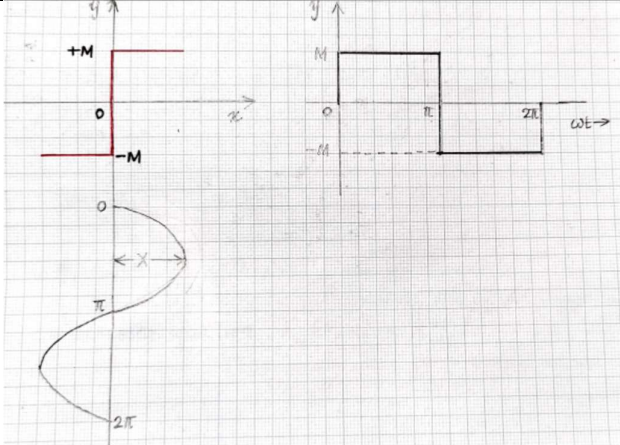
Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
SEVENTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2022			
Course Code: EET401			
Course Name: ADVANCED CONTROL SYSTEMS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions, each carries 3 marks.	Marks
1		(i) state variables (1mark) (ii) phase variables (1mark) and (iii) state space (1mark)	(3)
2		Select state variables and output variable, substitute them in the basic Kirchoff's equations and rearrange the terms. (1 mark) $v(t) = Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t)dt = Ri(t) + L \frac{di(t)}{dt} + v_c(t)$ Select voltage across capacitor as one state variable and inductor current (current in the series circuit) as the other state variable. Rearrange the terms. Final state equation (1 mark) $\dot{v}_o(t) = Ri(t)$ Substitute for $i(t)$ in terms of state variable. Final output equation (1 mark)	(3)
3		Statement of property and proof 1 mark for each property	(3)
4		$m = 5$ Method with steps 2 marks and 1 mark for final answer	(3)
5		Duality principle- Explanation with necessary equations (3 marks)	(3)
6		Explanation with necessary equations: New state model (2 marks) and condition (1mark) $\dot{Z} = \Lambda Z + \tilde{B}U$ where $\Lambda = M^{-1}AM$ & $\tilde{B} = M^{-1}B$ (2 marks) Condition: $\tilde{B}$ must have no rows with all zeros (1mark)	(3)
7		Any three characteristics (without explanation) (1 mark each) Jump resonance, Limit cycle, Frequency entrainment, Sub-harmonic oscillations, etc.	(3)
8		(i) jump resonance (1mark) (ii) stable limit cycle (1mark) and (iii) unstable limit cycle (1mark)	(3)
9		(i) singular point (1mark) (ii) vortex point (1mark) and (iii) saddle point (1mark)	(3)

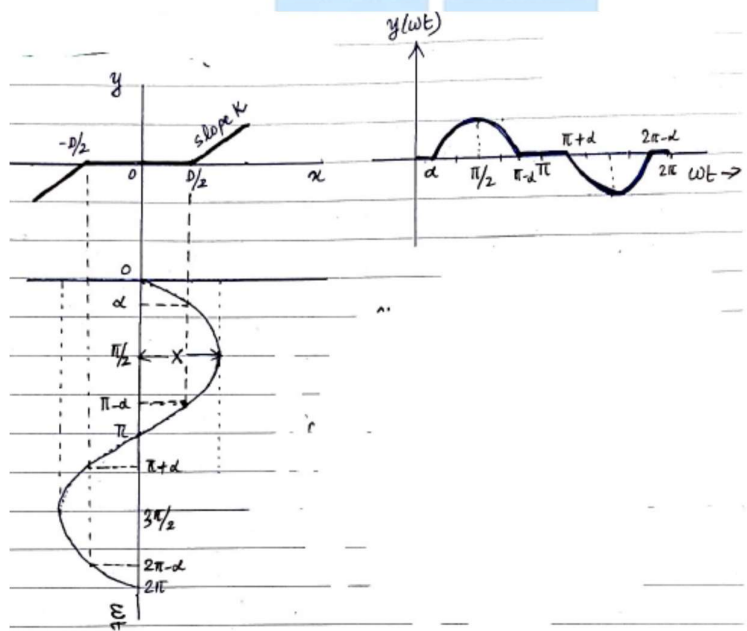
10		(i) positive definite & example (1 mark) (ii) negative definite & example (1 mark) and (iii) indefinite & example (1 mark)	(3)
PART B			
<i>Answer one full question from each module, each carries 14 marks.</i>			
		Module I	
11	a)	State model of an armature controlled dc motor: KVL equation for armature circuit: (1 mark) $V_a = I_a \cdot R_a + L_a \cdot \frac{di_a}{dt} + E_b \quad \left(E_b \propto \frac{d\theta}{dt} \right)$ Torque equation: $T \propto I_a$ ($T = K_T I_a$) (1 mark) Eqn. for electro mechanical system: $J \frac{d^2\theta}{dt^2} + B \frac{d\theta}{dt} = T$ (1 mark) Select suitable state variables and output variables, substitute them in the above three equations, rearrange the terms. (2 marks) Final state equation (1 mark) Final output equation (1 mark)	(7)
	b)	State model using phase variables $x_1 = y$ (1 mark) $x_2 = \frac{dy}{dt} = \dot{x}_1$ (1 mark) $x_3 = \frac{d^2y}{dt^2} = \dot{x}_2$ (1 mark) $\therefore \frac{d^3y}{dt^3} + 6 \frac{d^2y}{dt^2} + 11 \frac{dy}{dt} + 6y + u = 0 \Rightarrow \dot{x}_3 = -6x_3 - 11x_2 - 6x_1 + u$ (2 marks) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ (2 marks)	(7)
OR			
12	a)	Similarity transformation: (3 marks) Process of transforming a square matrix to another similar matrix by a transformation. Similarity transformation in state space analysis: (2 marks) The helpfulness of similarity transformation are, a. Diagonalization of a square matrix - Diagonal Canonical Form	(5)

	b. If we transform the system to controllable canonical form, state feedback controller design is easier c. If we transform the system to observable canonical form, observer design is easier d. If we transform the system to Diagonal/Jordan canonical form controllability, observability and stability can be easily identified, solution (STM) can be computed easily Any one point from the above is sufficient.	
	b) Canonical form $[\lambda I - A] = \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{bmatrix} \quad \lambda_1 = -1 \quad \lambda_2 = -2 \quad \lambda_3 = -3$ (3 marks) $M = [m_1 \quad m_2 \quad m_3] = \begin{bmatrix} 6 & 3 & 2 \\ -6 & -6 & -6 \\ 6 & 12 & 18 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$ (3 marks) $A_{new} = M^{-1} \cdot A \cdot M = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \quad B_{new} = M^{-1} \cdot B = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ (3 marks) $C_{new} = C \cdot M = [1 \quad 1 \quad 1]$	(9)
	Module II	
13	$z^2 Y(z) + 5zY(z) + 6Y(z) = U(z) \quad \therefore \frac{Y(z)}{U(z)} = \frac{1}{(z+3)(z+2)}$ (2 marks) On solving, $\frac{Y(z)}{U(z)} = \frac{-1}{(z+3)} + \frac{1}{(z+2)}$ (4 marks) $x_1(k+1) = -3x_1(k) + u(k)$ $x_2(k+1) = -2x_2(k) + u(k)$ $y(k) = -x_1(k) + x_2(k)$ $\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} u(k) \\ u(k) \end{bmatrix}$ 50% credit for steps $y(k) = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$ (6 marks) Block diagram (1 mark)	(14)

		STM: $A^k = Z^{-1} \left\{ [zI - A]^{-1} \cdot z \right\}$ $= Z^{-1} \left\{ \begin{bmatrix} \frac{1}{(z+3)} & 0 \\ 0 & \frac{1}{(z+2)} \end{bmatrix} z \right\}$ $= \begin{bmatrix} -3^k & 0 \\ 0 & -2^k \end{bmatrix}$ <i>(1 mark)</i> <i>50% credit for steps</i> The STM can be directly written without above computation also.	
		OR	
14	a)	STM using Laplace transform method $[sI - A] = \begin{bmatrix} (s-1) & 0 \\ -1 & (s-1) \end{bmatrix}$ $STM = e^{At} = L^{-1} \left\{ [sI - A]^{-1} \right\}$ $= \begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix}$ $X(t) = e^{At} \cdot X(0)$ $= \begin{bmatrix} e^t \\ te^t \end{bmatrix}$ <i>(1 mark)</i> <i>(1 mark)</i> <i>(3 marks)</i> <i>(1 mark)</i> <i>(1 mark)</i>	(7)
	b)	STM using Cayley-Hamilton Theorem $ \lambda I - A = \begin{vmatrix} \lambda & -1 \\ 2 & \lambda + 3 \end{vmatrix} \quad \lambda_1 = -1 \text{ \& } \lambda_2 = -2$ $f(\lambda_1) = \alpha_0 + \alpha_1 \cdot \lambda_1 = e^{\lambda_1 t} = e^{-t}$ $f(\lambda_2) = \alpha_0 + \alpha_1 \cdot \lambda_2 = e^{\lambda_2 t} = e^{-2t}$ $\therefore \alpha_0 = 2e^{-t} - e^{-2t} \quad \& \quad \alpha_1 = e^{-t} - e^{-2t}$ $f(A) = e^{At} = \alpha_0 \cdot I + \alpha_1 \cdot A = \begin{bmatrix} (2e^{-t} - e^{-2t}) & (e^{-t} - e^{-2t}) \\ (-2e^{-t} + 2e^{-2t}) & (-e^{-t} + 2e^{-2t}) \end{bmatrix}$ <i>(1 mark)</i> <i>(1 mark)</i> <i>(1 mark)</i> <i>(2 marks)</i> <i>(2 marks)</i>	(7)
		Module III	
15	a)	Kalman's method for observability Composite matrix : $Q_0 = \begin{bmatrix} C^T & A^T \cdot C^T & (A^T)^2 \cdot C^T \end{bmatrix}$ <i>(2 marks)</i>	(6)

		$Q_0 = \begin{bmatrix} 40 & 0 & 0 \\ 10 & 40 & -30 \\ 0 & 10 & 0 \end{bmatrix}$ <i>50% credit for steps</i> (3 marks) $Q_0 \neq 0$ System is completely observable (1 mark)	
	b)	<i>There is no method specified for design in the question. Examiners are requested to verify the method applied and award the marks accordingly. 7 marks for steps.</i> Observer gain matrix $G = \begin{bmatrix} 25.2 \\ 21.5 \\ 10 \end{bmatrix}$ (1 mark)	(8)
		OR	
16	a)	PBH method for controllability : condition $\text{rank} \left\{ \begin{bmatrix} (\lambda I - A) & B \end{bmatrix} \right\} = n$ (1 mark) $n = 2 \quad \lambda_1 = -1 \quad \lambda_2 = -4$ (2 marks) For λ_1 : $\text{rank} \left\{ \begin{bmatrix} -1 & -1 & 0 \\ 4 & 4 & 1 \end{bmatrix} \right\} = 2$ (1 mark) For λ_2 : $\text{rank} \left\{ \begin{bmatrix} -4 & -1 & 0 \\ 4 & 1 & 1 \end{bmatrix} \right\} = 2$ (1 mark) System is completely controllable. (1 mark)	(6)
	b)	Feed back controller with a state feed back: <i>There is no method specified for design in the question. Examiners are requested to verify the method applied and award the marks accordingly. 7 marks for steps.</i> State feedback gain matrix: $K = [0.4 \quad 0.4 \quad 0.1]$ (1 mark)	(8)
		Module IV	
17	a)	Describing function of ideal relay non-linearity: Waveforms :	(6)

		 Input signal: $x = X \sin \omega t$ Output signal: $y(\omega t) = \begin{cases} +M & ; 0 < \omega t < \pi \\ -M & ; \pi < \omega t < 2\pi \end{cases}$ Describing function : $A_1 = \frac{4M}{\pi} \quad B_1 = 0 \quad \phi_1 = 0^\circ$ Derivation of constants with necessary steps $K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \quad Y_1 = \sqrt{A_1^2 + B_1^2} \quad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right)$ $K_N(X, \omega) = \frac{4M}{\pi X} \angle 0^\circ$	(1 mark)	
	b)	Frequency and nature of the limit cycle: Plot of $G(j\omega)$ $\frac{-1}{K_N(X, \omega)}$ locus $G(j\omega) = \frac{10}{s(s+1)(s+2)} = \frac{5}{j\omega(1+j\omega)(1+j0.5\omega)}$ $ G(j\omega) = \frac{5}{\omega \sqrt{1+\omega^2} \sqrt{1+0.25\omega^2}} \quad \angle G(j\omega) = -90^\circ - \tan^{-1}(\omega) - \tan^{-1}(0.5\omega)$ $\frac{-1}{K_N(X, \omega)} = \frac{\pi X}{4} \angle -180^\circ$	(2 marks) (2 marks)	(8)

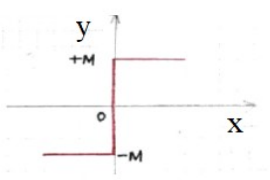
		<table border="1"> <tr> <td>ω rad/s</td><td>.....</td><td>.....</td><td>.....</td></tr> <tr> <td>$G(j\omega)$</td><td>.....</td><td>.....</td><td>.....</td></tr> <tr> <td>$\angle G(j\omega)$</td><td>.....</td><td>-180°</td><td>.....</td></tr> </table>	ω rad/s				$ G(j\omega) $				$\angle G(j\omega)$		-180°		
ω rad/s															
$ G(j\omega) $															
$\angle G(j\omega)$		-180°													
		$G(j\omega)$ locus and $\frac{-1}{K_N(X, \omega)}$ locus intersect at -180° axis (1 mark) $-90 - \tan^{-1}(\omega) - \tan^{-1}(0.5\omega) = -180^\circ$ (2 marks) $\therefore \omega = \sqrt{2} \text{ rad/s}$ Nature of the limit cycle: stable (1 mark) <i>Student may assume the shape of $G(j\omega)$ as it crosses the other locus at -180° axis. In that case, they may only need to calculate the point of intersection.</i>													
		OR													
18	a)	Describing function of dead-zone non-linearity: Waveforms :  Input signal: $x = X \sin \omega t$ Output signal: $y(\omega t) = \begin{cases} 0 & ; 0 < \omega t < \alpha \\ K \left(x - \frac{D}{2} \right) & ; \alpha < \omega t < (\pi - \alpha) \\ 0 & ; (\pi - \alpha) < \omega t < \pi \end{cases}$	(6)												

	Describing function : $K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \qquad Y_1 = \sqrt{(A_1^2 + B_1^2)} \qquad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right) \qquad (2 \text{ marks})$ $A_1 = \frac{4K}{\pi} \left[\frac{X}{2} \left(\frac{\pi}{2} - \alpha + \frac{\sin 2\alpha}{2} \right) - \frac{D}{2} \cos \alpha \right] = KX \left[1 - \frac{2}{\pi} (\alpha + \sin \alpha \cdot \cos \alpha) \right]$ $B_1 = 0 \qquad \phi_1 = 0^0$ Derivation of constants with necessary steps (final answer either in terms of D or without D) (2 marks) <i>Alternate expression for A_1: $A_1 = \frac{KX}{\pi} \{\pi - 2\alpha - \sin 2\alpha\}$</i>												
b)	Frequency and nature of the limit cycle: (8) Plot of $G(j\omega)$ (2 marks) $\frac{-1}{K_N(X, \omega)}$ locus (2 marks) $G(j\omega) = \frac{1}{j\omega(1+j2\omega)(1+j\omega)}$ $ G(j\omega) = \frac{1}{\omega \sqrt{(1+4\omega^2)} \sqrt{(1+\omega^2)}} \qquad \angle G(j\omega) = -90^0 - \tan^{-1}(2\omega) - \tan^{-1}(\omega)$ $\frac{-1}{K_N(X, \omega)} = \begin{cases} \frac{\pi}{2K\pi[\beta + \sin \beta \cdot \cos \beta]} \angle -180^0 & ; X > 1 \\ \frac{1}{K} \angle -180^0 & ; X < 1 \end{cases}$ <table><tr><td>ω rad/s</td><td>.....</td><td>0.707</td><td>.....</td></tr><tr><td>$G(j\omega)$</td><td>.....</td><td>.....</td><td>.....</td></tr><tr><td>$\angle G(j\omega)$</td><td>.....</td><td>-180⁰</td><td>.....</td></tr></table> $G(j\omega)$ locus and $\frac{-1}{K_N(X, \omega)}$ locus intersect at -180⁰ axis (1 mark) $-90 - \tan^{-1}(2\omega) - \tan^{-1}(\omega) = -180^0$ $\therefore \omega = 0.707 \text{rad / s}$ (2 marks) Nature of the limit cycle: stable (1 mark) <i>Student may assume the shape of $G(j\omega)$ as it crosses the other locus at -180⁰ axis. In that case, they may only need to calculate the point of intersection.</i>	ω rad/s		0.707		$ G(j\omega) $				$\angle G(j\omega)$		-180 ⁰	
ω rad/s		0.707											
$ G(j\omega) $													
$\angle G(j\omega)$		-180 ⁰											

		Module V	
19	a)	Phase trajectory (1 mark) Procedure to draw the phase trajectory with necessary sketches (6 marks)	(7)
	b)	Lyapunov function: $-Q = A^T \cdot P + P \cdot A$ $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix}^T \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix} \quad (p_{12} = p_{21})$ (1 mark) On solving, $P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = \begin{bmatrix} 23/60 & -7/60 \\ -7/60 & 11/60 \end{bmatrix}$ (3 marks) Positive definiteness of P for stability. $ p_{11} > 0$ $\begin{vmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{vmatrix} = \frac{17}{300} > 0$ (1 mark) Lyapunov function: $V(X) = X^T \cdot P \cdot X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{23}{60} x_1^2 - \frac{14}{60} x_1 \cdot x_2 + \frac{11}{60} x_2^2$ (2 marks)	(7)
		OR	
20	a)	Plot of phase trajectory (with necessary steps) (7 marks) full marks may be awarded if the shape of trajectory is identified with atleast 3 points marked and joined by a smooth curve (including the initial point).	(7)
	b)	Lyapunov function (1 mark) Condition/s for the system to be stable (1 mark) Condition/s for the system to be asymptotically stable in large (1 mark) Condition/s for the system to be unstable (1 mark)	(7)
		$\frac{dV(X)}{dt} = \left(\frac{\partial V}{\partial x_1} \right) \dot{x}_1 + \left(\frac{\partial V}{\partial x_2} \right) \dot{x}_2 = -2x_2^4$ (2 marks) System is stable (1 mark)	

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Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
SEVENTH SEMESTER B. TECH (S) DEGREE EXAMINATION, MAY 2023			
Course Code: EET401			
Course Name: ADVANCED CONTROL SYSTEMS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions, each carries 3 marks.	Marks
1		Any three advantages (1Mark Each)	(3)
2		Definition of Phase variable (1Mark), Any two advantages (1Mark Each)	(3)
3		Complete derivation (3 Marks) $\dot{X}=AX +BU, Y=CX+DU$ Taking Laplace Transform and initial conditions where zero $SX(S)=AX(S)+BU(S),$ $X(S)=[(SI-A)^{-1}B] U(S)$ $Y(S)=CX(S)+DU(S)=[C(SI-A)^{-1}B+D] U(S)$ $Y(S)/U(S)= [C(SI-A)^{-1}B+D]$	(3)
4		Three properties (1Mark Each)	(3)
5		Definition of Controllability(1Mark) Steps for controllability check using Kalman's test (2 Marks)	(3)
6		Duality principle- Explanation with necessary equations(3Marks)	(3)
7		Frequency entrainment explanation (2 Marks), Characteristic Curve (1 Mark)	(3)
8		Input signal: $x = X \cdot \sin \omega t$ Output signal: $y(\omega t) = \begin{cases} +M & ; 0 < \omega t < \pi \\ -M & ; \pi < \omega t < 2\pi \end{cases}$  Describing function : $K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \quad Y_1 = \sqrt{(A_1^2 + B_1^2)} \quad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right)$ $A_1 = \frac{4M}{\pi} \quad B_1 = 0 \quad \phi_1 = 0^0$	(3)
9		Explanation with diagrams (3Marks)	(3)

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10	The middle matrix is found as the coeff. of $\begin{bmatrix} x_1^2 & \frac{x_1x_2}{2} & \frac{x_1x_3}{2} \\ \frac{x_2x_1}{2} & x_2^2 & \frac{x_2x_3}{2} \\ \frac{x_1x_3}{2} & \frac{x_2x_3}{2} & x_3^2 \end{bmatrix}$ (1 Mark) $x_1^2 + 3x_2^2 + 11x_3^2 - 2x_1x_2 + 4x_2x_3 + 2x_1x_3 = [x_1 \ x_2 \ x_3] \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & 2 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ Now, $p_{11} = 1 > 0$ $\begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} = 2 > 0$ $\begin{vmatrix} 1 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & 2 & 11 \end{vmatrix} = 11 > 0$ (1 Mark) Quadratic form is positive definite (1 Mark)	(3)
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PART B

Answer one full question from each module, each carries 14 marks.

	Module I	
11	a) $x_1 = i_L$, $x_2 = v_c$ (1 Mark) $\frac{L di_L}{dt} + i_L R_2 = v_c$ $\dot{x}_1 = \frac{-10}{5} x_1 + \frac{v_c}{5}$ (2 Marks) $i_L + C \frac{dv_c}{dt} = \frac{v_0 - v_c}{R_1}$ $\dot{x}_2 = \frac{(v_0 - v_c)5}{5} - 5x_1$ (2 Marks) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 1/5 \\ -5 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$ (2Marks) b) State model of field-controlled dc motor: KVL equation for field circuit $v_f = R_f i_f + L_f \frac{di_f}{dt}$ (1 mark) Torque equation $T \propto i_f$, $T = K_{tf} i_f$ (1 mark) Equation of electro mechanical system: $J \frac{d^2\theta}{dt^2} + B \frac{d\theta}{dt} = K_{tf} i_f$ (1 mark) Selection of state variables and output variables. (1 mark) Substitution and rearrangement (1mark) Final state equation (1 mark) Final output equation (1 mark)	(7)
	OR	

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12	a)	$x_1 = y, x_2 = \dot{y} = \dot{x}_1, x_3 = \ddot{y} = \dot{x}_2$ (1 mark) $\dot{x}_1 = x_2$ (1 mark) $\dot{x}_2 = x_3$ (1 mark) $\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 10y = 8u(t)$. i.e. $\dot{x}_3 = 8u(t) - 10x_1 - 11x_2 - 6x_3$ (2 mark) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -10 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 8 \end{bmatrix} u$ (1mark) $y = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ (1 mark)	(7)
	b)	$\frac{Y(s)}{U(s)} = \frac{2(s+5)}{(s+2)(s+3)(s+4)} = \frac{A}{(s+2)} + \frac{B}{(s+3)} + \frac{C}{(s+4)}$ (1Mark) A=3, B= -4, C=1 (3Marks) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$ (2Marks) $y = [3 \ -4 \ 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ (1Mark)	(7)
Module II			
13	a)	$(sI - A) = \begin{bmatrix} s & -1 \\ 6 & s+5 \end{bmatrix}$ (1 Mark) $e^{At} = L^{-1} [(sI - A)^{-1}] = \begin{bmatrix} 3e^{-2t} - 2e^{-3t} & e^{-2t} - e^{-3t} \\ -6e^{-2t} + 6e^{-3t} & -2e^{-2t} + 3e^{-3t} \end{bmatrix}$ (4 Marks) $X(t) = e^{At} X(0) = \begin{bmatrix} 3e^{-2t} - 2e^{-3t} & \\ -6e^{-2t} + 6e^{-3t} & \end{bmatrix}$ (2 Marks)	(7)
	b)	$\frac{Y(z)}{U(z)} = \frac{4z^3 - 12z^2 + 13z - 7}{z^3 - 4z^2 + 5z - 2}$	(7)

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		$x_1(k+1) = x_2$ $x_2(k+1) = x_3$ $x_3(k+1) = 2x_1 - 5x_2 + 4x_3 + u \text{ (2Marks)}$ $y = -7x_1 + 13x_2 - 12x_3 + 4x_3(k+1)$ $y = x_1 - 7x_2 + 4x_3 + 4u \text{ (2Marks)}$ $\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \text{ (2Marks)}$ $y = [1 \quad -7 \quad 4] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 4u \text{ (1 mark)}$	
		OR	
14	a)	$[\lambda I - A] = \begin{bmatrix} \lambda & -1 \\ 2 & \lambda + 3 \end{bmatrix} \text{ (1 Mark)}$ $ \lambda I - A = 0, \lambda_1 = -1, \lambda_2 = -2 \text{ (1 Mark)}$ $\alpha_0 + \alpha_1 \lambda_1 = e^{\lambda_1 t} = e^{-t}$ $\alpha_0 + \alpha_1 \lambda_2 = e^{\lambda_2 t} = e^{-2t} \text{ (2 Mark)}$ $\alpha_1 = e^{-t} - e^{-2t}$ $\alpha_0 = 2e^{-t} - e^{-2t} \text{ (2 Mark)}$ $e^{At} = \alpha_0 I + \alpha_1 A \text{ (1 Mark)}$ $e^{At} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -2e^{-t} + 2e^{-2t} \end{bmatrix} \text{ (1 Mark)}$	(8)
	b)	$T(s) = C(SI - A)^{-1}B + D \text{ (1 Mark)}$ $(sI - A) = \begin{bmatrix} s+3 & -1 \\ 2 & s \end{bmatrix} \text{ (1 Mark)}$ $(SI - A)^{-1} = \begin{bmatrix} \frac{s}{s^2+3s+2} & \frac{1}{s^2+3s+2} \\ \frac{-2}{s^2+3s+2} & \frac{s+3}{s^2+3s+2} \end{bmatrix} \text{ (2 Marks)}$ $T(s) = [1 \quad 0] \begin{bmatrix} \frac{s}{s^2+3s+2} & \frac{1}{s^2+3s+2} \\ \frac{-2}{s^2+3s+2} & \frac{s+3}{s^2+3s+2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ (1 Mark)}$ $T(S) = \frac{1}{s^2+3s+2} \text{ (1 Mark)}$	(6)
		Module III	
15	a)	Necessary and sufficient condition for complete observability is none of the columns of $\tilde{C}$ matrix is zero $\tilde{C} = CM$ (1 Mark)	(10)

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		$ \lambda I - A = 0, \lambda_1 = -1, \lambda_2 = -1, \lambda_3 = -4$ (2 Marks) $M = \begin{bmatrix} 4 & 4 & 1 \\ -4 & -2 & 2 \\ -4 & 0 & -4 \end{bmatrix}$ (4 Marks) $\tilde{C} = CM = [1 \ 0 \ 0] \begin{bmatrix} 4 & 4 & 1 \\ -4 & -2 & 2 \\ -4 & 0 & -4 \end{bmatrix} = [4 \ 4 \ 1]$ (2 Marks) None of the columns of $\tilde{C}$ matrix is zero. Hence the system is completely observable. (1 Mark)	
	b)	Explanation of Full order observer(2Marks), Reduced order observer(2Marks)	(4)
		OR	
16	a)	$\lambda_1 = -1, \lambda_2 = -3$ $\text{Rank of } [(\lambda I - A) \ B] = 2$ (full rank) for both λ_1 and λ_2 Hence system is completely controllable (4 Marks) $\text{Rank of } \begin{bmatrix} C \\ (\lambda I - A) \end{bmatrix} = 2$ (full rank) for both λ_1 and λ_2 Hence system is completely observable (4 Marks)	(8)
	b)	Pole placement technique explanation (6Marks)	(6)
		Module IV	
17	a)	Definition of describing function (2 Marks) Stability Analysis- Concept of Enclosure of $-1/K_N$ locus by $G(j\omega)$ locus with necessary diagrams (2 Marks) Occurrence of Limit Cycle (2Marks)	(6)
	b)	Derivation of describing function of relay with dead zone with all necessary steps Waveforms(2Mark) Output Equation (1 Mark) Derivation (3 Marks) Final Answer (2 Marks)	(8)
		OR	
18	a)	Explanation of any three non-linearities with waveforms (2 Marks each)	(6)
	b)	Derivation of describing function of saturation nonlinearity with necessary steps Waveforms(2Mark) Output Equation (1 Mark)	(8)

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		Derivation (3 Marks) Final Answer (2 Marks)	
		Module V	
19	a)	Definition of Singular point (1 Mark), Any five types of singular point with nature of eigen values or phase portrait diagram (1 Mark each)	(6)
	b)	Lyapunovs Stability Criterion (3 Marks) $\dot{x}_1 = -3x_1 + x_2$ $\dot{x}_2 = -x_1 - x_2 - x_2^3$ $V(x) = x_1^2 + x_2^2$ $\dot{V}(x) = \frac{\partial V}{\partial x_1} \dot{x}_1 + \frac{\partial V}{\partial x_2} \dot{x}_2$ $\dot{V}(x) = 2x_1 \dot{x}_1 + 2x_2 \dot{x}_2 \quad (2 \text{ Marks})$ Substitute the values of $\dot{x}_1$ and $\dot{x}_2$ $\dot{V}(x) = 2x_1(-3x_1 + x_2) + 2x_2(-x_1 - x_2 - x_2^3)$ $\dot{V}(x) = -6x_1^2 - 2x_2^2 - 2x_2^4 = \text{negative definite} \quad (2 \text{ Marks})$ System is asymptotically stable in the large. (1 Mark)	(8)
		OR	
20	a)	$A^T P + P A = -Q$ $\begin{bmatrix} -1 & 1 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \quad (2 \text{ Marks})$ Solve for matrix P $P = \begin{bmatrix} 23/60 & -7/60 \\ -7/60 & 11/60 \end{bmatrix} \quad (5 \text{ Marks})$ Applying Sylvester's theorem prove that matrix P is positive definite. (2 Marks) Equilibrium state is stable (1 Mark)	(10)
	b)	Definition of Stability (1 Mark), Asymptotic stability (1 Mark), Asymptotically stable in the large (1 Mark), Instability. (1 Mark),	(4)

Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY		
SEVENTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2021		
Course Code: EET401		
Course Name: Advanced Control Systems		
PART A		
Answer all questions. Each question carries 3 marks		
	Q1	What do you mean by (i) state variables (ii) phase variables and (iii) state space?
Ans		State variables are a set of measurable quantities or variables that describe the complete state of a physical system at any given time. They are the minimal set of variables required to completely characterize the system. For example, in a simple mechanical system, the position and velocity of the object are the state variables. Phase variables are a subset of state variables that can be used to describe the behavior of a dynamical system. They are chosen based on their ability to simplify the equations of motion for the system and make it easier to analyze. Phase variables can be expressed as functions of the state variables. (iii) State space is a mathematical concept used to describe the collection of all possible states of a system. It is represented as a multi-dimensional space, where each dimension corresponds to a different state variable. The state of the system at any given time can be represented by a point in this space. The behavior of the system over time can be described as a trajectory or path through the state space. State space analysis is a powerful tool used in the study of dynamical systems.
	Q2	A series RLC circuit is excited by a voltage source, $v(t)$ volts and the output is measured across the resistor. Derive the state model of the electrical system
Ans		To derive the state model of a series RLC circuit excited by a voltage source, we need to first write the differential equations governing the circuit. Let's assume that the voltage source has a time-varying voltage of $v(t)$ volts, and the circuit consists of a resistor of resistance R ohms, an inductor of inductance L henries, and a capacitor of capacitance C farads, connected in series. Applying Kirchhoff's voltage law to the circuit, we can write:

	$v(t) = L \frac{di(t)}{dt} + Ri(t) + \frac{1}{C} \int i(t) dt$ where $i(t)$ is the current flowing through the circuit at time t. This is a second-order differential equation, which can be written as a system of first-order differential equations by introducing two state variables: $x_1(t) = i(t)$ (current through the circuit) $x_2(t) = \frac{di(t)}{dt}$ (rate of change of current) Differentiating $x_1(t)$ with respect to time, we get: $\frac{dx_1}{dt} = \frac{di}{dt} = x_2$ Substituting these equations in the original differential equation, we get: $v(t) = L \frac{dx_2}{dt} + Rx_1 + \frac{1}{C} \int x_1(t) dt$ This gives us a set of two coupled first-order differential equations: $\frac{dx_1}{dt} = x_2$ $\frac{dx_2}{dt} = (-R/L)x_2 - (1/L)x_1 + (1/L)v(t)$ This is the state model of the series RLC circuit excited by a voltage source, where x_1 and x_2 are the state variables, and $v(t)$ is the input.
Q3	State and prove any three properties of state transition matrix
Ans	The state transition matrix, also known as the transition matrix or the system matrix, is a matrix that describes the evolution of the state of a dynamical system over time. It is a fundamental concept in the theory of linear dynamical systems and has several important properties. Here are three properties of the state transition matrix and their proofs: Homogeneity: The state transition matrix is homogeneous, which means that it is time-invariant. This property can be stated mathematically as follows: $\Phi(t+s, t) = \Phi(t+s, s) \Phi(s, t)$ where $\Phi(t+s, t)$ is the state transition matrix from time t to time $t+s$, and $\Phi(t+s, s)$ and $\Phi(s, t)$ are the state transition matrices from time s to time $t+s$ and from time t to time s, respectively. Proof: Let $x(t)$ be the state vector of the system at time t. Then, we have: $x(t+s) = \Phi(t+s, t) x(t)$ $x(t+s) = \Phi(t+s, s) \Phi(s, t) x(t)$ (by the definition of the state transition matrix)

	Therefore, $\Phi(t+s, t) = \Phi(t+s, s) \Phi(s, t)$, which proves the homogeneity property. Semi-group property: The state transition matrix has the semi-group property, which means that the matrix product of two state transition matrices is also a state transition matrix. This property can be stated mathematically as follows: $\Phi(t_3, t_1) = \Phi(t_3, t_2) \Phi(t_2, t_1)$ where $t_1 \leq t_2 \leq t_3$. Proof: Let $x(t)$ be the state vector of the system at time t. Then, we have: $x(t_3) = \Phi(t_3, t_1) x(t_1)$ $x(t_3) = \Phi(t_3, t_2) \Phi(t_2, t_1) x(t_1) \text{ (by the definition of the state transition matrix)}$ Therefore, $\Phi(t_3, t_1) = \Phi(t_3, t_2) \Phi(t_2, t_1)$, which proves the semi-group property. Invertibility: The state transition matrix is invertible, which means that there exists a unique inverse matrix that describes the evolution of the system in the reverse direction. This property can be stated mathematically as follows: $\Phi(t, t_0)^{-1} = \Phi(t_0, t)$ Proof: Let $x(t)$ be the state vector of the system at time t. Then, we have: $x(t) = \Phi(t, t_0) x(t_0)$ $x(t_0) = \Phi(t_0, t) x(t)$ Therefore, $\Phi(t, t_0)^{-1} = \Phi(t_0, t)$, which proves the invertibility property.
Q4	The state space representation and transfer function of a system are given below. Find the value of 'm'. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ $\frac{Y(s)}{U(s)} = \frac{2s+2}{(s+1)(s+4)}$
Ans	To find the value of 'm', we need to use the relationship between the state space representation and the transfer function of a system. The transfer function of a system is the

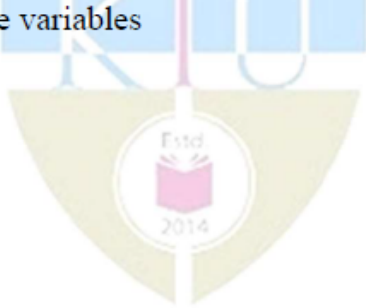
	Laplace transform of its output with respect to its input, when all initial conditions are zero. The transfer function is given by: $H(s) = C(sI - A)^{-1}B + D$ where A, B, C, and D are the matrices that define the state space representation of the system, and I is the identity matrix of appropriate size. From the given state space representation, we have: $A = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 2m & 4 \\ 2 & 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \quad D = 0$ Substituting these values in the transfer function equation, we get: $H(s) = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} (sI - A)^{-1} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ We can evaluate the inverse of (sI - A) using the standard formula for the inverse of a 3x3 matrix. The resulting expression for H(s) is quite long and complicated, so we won't write it out explicitly here. However, we can use the fact that the transfer function is a rational function of s, and find the value of 'm' by equating the coefficients of s² in the numerator and denominator of H(s). Setting the coefficient of s² in the numerator of H(s) to zero, we get: $m = 1$ Therefore, the value of 'm' is 1. $Ri(t) + L \frac{d}{dt} i(t) = V$ $R I(s) + L s I(s) = \frac{V}{s}$ $I(s) = \frac{\left(\frac{V}{R}\right)}{s} + \frac{\left(-\frac{V}{R}\right)}{\left(s + \frac{R}{L}\right)}$ $i(t) = \frac{V}{R} \left(1 - e^{-\left(\frac{Rt}{L}\right)} \right)$
Q5	With necessary equations, explain duality principle .

	The duality principle is a powerful tool in control system theory that allows us to simplify the analysis of a system by interchanging the roles of input and output variables, and poles and zeros of the transfer function. The resulting dual system preserves certain properties of the original system, such as relative degree and stability margins, and can provide useful insights into the behavior of the system.
Q6	Explain the Gilbert's method to test controllability of the n^{th} order system represented by the state model given below. Assume that the eigen values are distinct. $\dot{X} = AX + BU \quad Y = CX + DU$
Ans	Gilbert's method is a technique used to test the controllability of a linear time-invariant (LTI) system represented in state-space form. The method is based on the concept of the controllability matrix, which is a square matrix that characterizes the controllability of the system. The controllability matrix is defined as: $Q_c = [B, AB, A^2B, \dots, A^{(n-1)}B]$ where A and B are the system matrices, and n is the order of the system. The controllability matrix Q_c has dimensions $(n \times m)$, where m is the number of inputs to the system. To apply Gilbert's method to test the controllability of an nth order system represented by the state model: $\dot{x} = Ax + Bu \quad y = Cx + Du$ where A is an $(n \times n)$ matrix, B is an $(n \times m)$ matrix, C is a $(p \times n)$ matrix, and D is a $(p \times m)$ matrix, we follow these steps: Step 1: Calculate the controllability matrix Q_c as defined above. Step 2: Compute the matrix M as follows: $M = [A^{(n-1)}B, A^{(n-2)}B, \dots, AB, B]$ Step 3: Compute the determinant of the matrix M. If the determinant is non-zero, then the system is controllable. If the determinant is zero, then the system is uncontrollable. In summary, Gilbert's method involves computing the controllability matrix and a related matrix M, and then testing the determinant of M to determine the controllability of the system. Note that this method assumes that the eigenvalues of the system matrix A are distinct. If the eigenvalues are not distinct, then additional analysis is required to test the controllability of the system.
Q7	Write any three characteristics of non-linear systems.

Ans	Nonlinear systems are systems that exhibit nonlinear relationships between their inputs and outputs. Here are three characteristics of nonlinear systems: Nonlinear systems may exhibit unexpected behaviors: Unlike linear systems, nonlinear systems may exhibit unexpected behaviors, such as oscillations, chaotic behavior, and bifurcations. These behaviors can arise due to the complex interactions between the system components and the nonlinearity of the system. Nonlinear systems may have multiple solutions: Nonlinear systems may have multiple solutions, unlike linear systems which typically have a unique solution. This means that the output of a nonlinear system can vary widely depending on the initial conditions and inputs. Nonlinear systems may have interactions between their components: In nonlinear systems, the components of the system can interact with each other in complex ways, making it difficult to analyze the behavior of the system. This is because the behavior of each component depends not only on its own state, but also on the states of other components. Overall, nonlinear systems are often more complex and difficult to analyze than linear systems, due to their nonlinear behavior and the interactions between their components.
Q8	Explain the following (i) jump resonance (ii) stable limit cycle and (iii) unstable limit cycle?
Ans	(i) Jump resonance: Jump resonance is a nonlinear phenomenon that occurs in systems with a periodic forcing function, where the amplitude of the response suddenly jumps to a higher value when the frequency of the forcing function matches the natural frequency of the system. This phenomenon can occur in systems with a nonlinear restoring force, such as systems with a hysteresis nonlinearity, and can result in large amplitude vibrations or failure of the system. (ii) Stable limit cycle: A stable limit cycle is a periodic solution of a nonlinear system that is attracting, meaning that if the system is perturbed slightly from the limit cycle, it will return to the limit cycle over time. This means that the system will oscillate with a fixed amplitude and frequency over time. Stable limit cycles can arise in systems with nonlinear dynamics, such as systems with a saturating nonlinearity or a negative damping term. (iii) Unstable limit cycle: An unstable limit cycle is a periodic solution of a nonlinear system that is repelling, meaning that if the system is perturbed slightly from the limit cycle, it will move away from the limit cycle over time. This means that the system will oscillate with a fixed amplitude and frequency over time, but the oscillations will grow over time rather than decay. Unstable limit cycles can arise in systems with nonlinear dynamics, such as systems with a positive damping term or a nonlinear feedback loop. In summary, jump resonance is a phenomenon where the amplitude of the response suddenly jumps to a higher value when the frequency of the forcing function matches the natural frequency of the system, stable limit cycles are periodic solutions of a nonlinear system that are attracting, and unstable limit cycles are periodic solutions of a nonlinear system that are repelling.
Q9	Define (i) singular point (ii) vortex point and (iii) saddle point

Ans	(i) Singular point: A singular point of a nonlinear system is a point in the state space where the dynamics of the system are not well-defined, meaning that the derivative of the system's state vector is either zero or undefined at that point. Singular points can be classified as either isolated or non-isolated, depending on whether there are nearby points with well-defined dynamics. (ii) Vortex point: A vortex point, also known as a center, is a type of singular point of a nonlinear system where the state trajectories form closed orbits around the point without spiraling inwards or outwards. This means that the state trajectories are periodic and the system's behavior is stable and repeatable over time. (iii) Saddle point: A saddle point, also known as a hyperbolic equilibrium or a saddle node, is a type of singular point of a nonlinear system where the state trajectories approach or diverge from the point in different directions. This means that the system's behavior is unstable and sensitive to small perturbations, and the point can act as a dividing line between different regions of the state space with different behaviors. In summary, singular points are points in the state space where the dynamics of the system are not well-defined, vortex points are singular points where the state trajectories form closed orbits around the point without spiraling inwards or outwards, and saddle points are singular points where the state trajectories approach or diverge from the point in different directions and the system's behavior is unstable and sensitive to small perturbations.
Q10	State the condition/s for a scalar function $V(x)$ to be (i) positive definite (ii) negative definite and (iii) indefinite. Give one example for each
Ans	A scalar function $V(x)$ is said to be: (i) Positive definite if $V(x) > 0$ for all $x \neq 0$, and $V(0) = 0$. In other words, the function is strictly positive for all non-zero values of x and zero at the origin. This implies that the system is stable in the sense that it converges to the origin, and any small perturbations from the equilibrium point decay over time. Example: $V(x) = x^2$ is positive definite. (ii) Negative definite if $V(x) < 0$ for all $x \neq 0$, and $V(0) = 0$. In other words, the function is strictly negative for all non-zero values of x and zero at the origin. This implies that the system is stable in the sense that it converges to the origin, but any small perturbations from the equilibrium point grow over time. Example: $V(x) = -x^2$ is negative definite. (iii) Indefinite if $V(x)$ takes both positive and negative values for different non-zero values of x, and $V(0) = 0$. In other words, the function can be both positive and negative for different values of x, and zero at the origin. This implies that the system is unstable and the behavior depends on the initial conditions. Example: $V(x) = x^3$ is indefinite.

	In summary, a scalar function $V(x)$ is positive definite if it is strictly positive for all non-zero values of x and zero at the origin, negative definite if it is strictly negative for all non-zero values of x and zero at the origin, and indefinite if it takes both positive and negative values for different non-zero values of x and zero at the origin.
PART B	
<i>Answer any one full question from each module, each carries 14 marks.</i> Module I	
Q11	a) From basics, derive the state model of an armature controlled dc motor. b) Differential equation of the system is given by, $\frac{d^3 y}{dt^3} + 6 \frac{d^2 y}{dt^2} + 11 \frac{dy}{dt} + 6y + u = 0$ where, y is the output and u is the input. Derive the state model of the system using phase variables
Ans	a) State model of an armature controlled dc motor: KVL equation for armature circuit: $V_a = I_a R_a + L_a \frac{di_a}{dt} + E_b \quad \left(E_b \propto \frac{d\theta}{dt} \right)$ Torque equation: $T \propto I_a \quad (T = K_T I_a)$ Eqn. for electro mechanical system: $J \frac{d^2 \theta}{dt^2} + B \frac{d\theta}{dt} = T$ Select suitable state variables and output variables, substitute them in the equations, rearrange the terms. Final state equation Final output equation

	b) State model using phase variables  $x_1 = y$ $x_2 = \frac{dy}{dt} = \dot{x}_1$ $x_3 = \frac{d^2y}{dt^2} = \dot{x}_2$ $\therefore \frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y + u = 0 \Rightarrow \dot{x}_3 = -6x_3 - 11x_2 - 6x_1 + u$ $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$
Q12	a) What is similarity transformation? How similarity transformation is helpful for state space analysis? Obtain the diagonal canonical form of the state model given below. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} u \quad [y] = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ b)
Ans	a) Similarity transformation is a linear transformation of a matrix in which the new matrix is obtained by multiplying the original matrix by an invertible matrix on both sides. Specifically, if A is a square matrix and P is an invertible matrix, then the similarity transformation of A by P is given by $P^{-1}AP$. Similarity transformations are useful in state space analysis because they help to identify the eigenvalues and eigenvectors of a system. The eigenvalues of a system are important

because they describe the stability and behavior of the system over time, and the eigenvectors correspond to the directions in which the system responds most strongly.

By performing a similarity transformation on the state space matrix A, it is possible to transform the system into a new coordinate system in which the new matrix A' is diagonal. The diagonal entries of A' are the eigenvalues of A, and the columns of the matrix P that is used to perform the transformation correspond to the eigenvectors of A. This diagonalization process is known as diagonalization by similarity transformation.

Once the state space matrix has been diagonalized, it is much easier to analyze the behavior of the system. In particular, the diagonal entries of the transformed matrix A' correspond to the decay rates of the system, and the eigenvectors of A' provide insight into the direction and strength of the system's response to different inputs. Overall, similarity transformations are a powerful tool for analyzing the behavior of complex systems described by state space models

b

Canonical form

$$[\lambda I - A] = \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{bmatrix} \quad \lambda_1 = -1 \quad \lambda_2 = -2 \quad \lambda_3 = -3$$

$$M = [m_1 \quad m_2 \quad m_3] = \begin{bmatrix} 6 & 3 & 2 \\ -6 & -6 & -6 \\ 6 & 12 & 18 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$$

$$A_{new} = M^{-1} \cdot A \cdot M = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \quad B_{new} = M^{-1} \cdot B = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$C_{new} = C \cdot M = [1 \quad 1 \quad 1]$$

Module 2

Q13	A discrete time system is given by, $y(k+2) + 5y(k+1) + 6y(k) = u(k)$ where, y is the output and u is the input. Assume zero initial conditions and the sampling period as 1 second. (i) Derive the state model of the system in Jordan canonical form (ii) Draw the block diagram of the state model and (iii) Compute the state transition matrix.
Ans	$z^2 Y(z) + 5zY(z) + 6Y(z) = U(z) \quad \therefore \frac{Y(z)}{U(z)} =$ On solving, $\frac{Y(z)}{U(z)} = \frac{-1}{(z+3)} + \frac{1}{(z+2)}$ $\begin{aligned} x_1(k+1) &= -3x_1(k) + u(k) \\ x_2(k+1) &= -2x_2(k) + u(k) \\ y(k) &= -x_1(k) + x_2(k) \end{aligned}$ $\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(k)$ $y(k) = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$ Block diagram $\begin{aligned} A^k &= Z^{-1} \left\{ [zI - A]^{-1} . z \right\} \\ &= Z^{-1} \left\{ \begin{bmatrix} \frac{1}{(z+3)} & 0 \\ 0 & \frac{1}{(z+2)} \end{bmatrix} z \right\} \\ &= \begin{bmatrix} -3^k & 0 \\ 0 & -2^k \end{bmatrix} \end{aligned}$
Q14	a) State equation of a system is given below. Compute the state transition matrix using Laplace transform method. Also, compute the solution of the given state equation if the initial state vector is $X(0)$. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

	b) Find the state transition matrix using Cayley-Hamilton theorem for the system matrix given below. $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$
Ans	a) STM using Laplace transform : $[sI - A] = \begin{bmatrix} (s-1) & 0 \\ -1 & (s-1) \end{bmatrix}$ $STM = e^{At} = L^{-1} \{ [sI - A]^{-1} \}$ $= \begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix}$ $X(t) = e^{At} \cdot X(0)$ $= \begin{bmatrix} e^t \\ te^t \end{bmatrix}$ b) STM using Cayley-Hamilton Theorem $ \lambda I - A = \begin{vmatrix} \lambda & -1 \\ 2 & \lambda + 3 \end{vmatrix} \quad \lambda_1 = -1 \text{ \& } \lambda_2 = -2$ $f(\lambda_1) = \alpha_0 + \alpha_1 \lambda_1 = e^{\lambda_1 t} = e^{-t}$ $f(\lambda_2) = \alpha_0 + \alpha_1 \lambda_2 = e^{\lambda_2 t} = e^{-2t}$ $\therefore \alpha_0 = 2e^{-t} - e^{-2t} \quad \& \quad \alpha_1 = e^{-t} - e^{-2t}$ $f(A) = e^{At} = \alpha_0 \cdot I + \alpha_1 \cdot A = \begin{bmatrix} (2e^{-t} - e^{-2t}) & (e^{-t} - e^{-2t}) \\ (-2e^{-t} + 2e^{-2t}) & (-e^{-t} + 2e^{-2t}) \end{bmatrix}$
Module 3	
Q15	

	a) Using Kalman's method, check the observability of the system given below. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad [y] = [40 \quad 10 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ b) Design the observer feed back gain matrix for the desired eigen values at -4, -3±j1 for the state model of the system given below. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 1 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} u \quad [y] = [0 \quad 0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$
Ans	a) Kalman's method for observability Composite matrix : $Q_0 = \begin{bmatrix} C^T & A^T.C^T & (A^T)^2.C^T \end{bmatrix}$ $Q_0 = \begin{bmatrix} 40 & 0 & 0 \\ 10 & 40 & -30 \\ 0 & 10 & 0 \end{bmatrix}$ b)

Observability matrix, $\mathcal{O}$:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CAA \end{bmatrix}$$

$$\mathcal{O} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 0 \\ 3 & 8 & 0 \\ 1 & 1 & 2 \\ 3 & 4 & 8 \\ 4 & 5 & 10 \\ 1 & 2 & 0 \\ 2 & 4 & 0 \end{bmatrix}$$

Rank of $\mathcal{O}$ is 3. Therefore, the system is observable.

$$s^3 + 10s^2 + 29s + 24 = (s + 4)(s + 3 + j)(s + 3 - j)$$

The observer pole locations are -4, -3+j, and -3-j.

The observer feedback gain matrix can be obtained using the formula:

$$L = \begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}^{-1} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix}$$

where

$$b_0 = 24$$

$$b_1 = 29 - 10L_1$$

$$b_2 = 10L_1 - L_1^2 - 10L_2 - 10L_3$$

Substituting the desired eigenvalues, we get:

	$b_0 = 24$ $b_1 = 101$ $b_2 = 262$ Solving for L, we get: $L = \begin{bmatrix} 4.9874 \\ -9.9487 \\ -3.0000 \end{bmatrix}$ Therefore, the observer feedback gain matrix $K = [k_1 \quad k_2 \quad k_3] = [L_1 \quad L_2 \quad L_3] =$ Observer gain matrix $G = \begin{bmatrix} 25.2 \\ 21.5 \\ 10 \end{bmatrix}$
Q16	a) Using PBH test, check the controllability of the system given below. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad [y] = [2 \quad 2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ b) Design a feed back controller with state feed back so that the closed loop poles are at -2, -1±j1. The state equation of the original systems is given below. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u$
Ans	a

PBH method for controllability : condition $\text{rank} \left\{ \begin{bmatrix} (\lambda I - A) & B \end{bmatrix} \right\} = n$

$$n = 2 \quad \lambda_1 = -1 \quad \lambda_2 = -4$$

For λ_1 : $\text{rank} \left\{ \begin{bmatrix} -1 & -1 & 0 \\ 4 & 4 & 1 \end{bmatrix} \right\} = 2$

For λ_2 : $\text{rank} \left\{ \begin{bmatrix} -4 & -1 & 0 \\ 4 & 1 & 1 \end{bmatrix} \right\} = 2$

System is completely controllable.

b

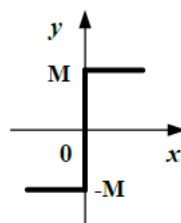
Feed back controller with a state feed back:

State feedback gain matrix: $K = [0.4 \quad 0.4 \quad 0.1]$

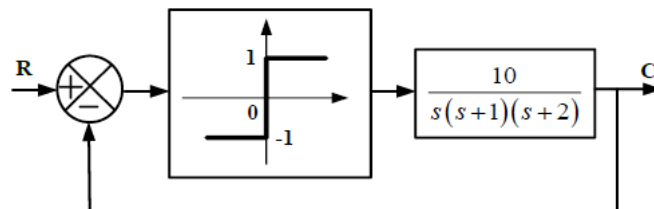
Module 4

Q17

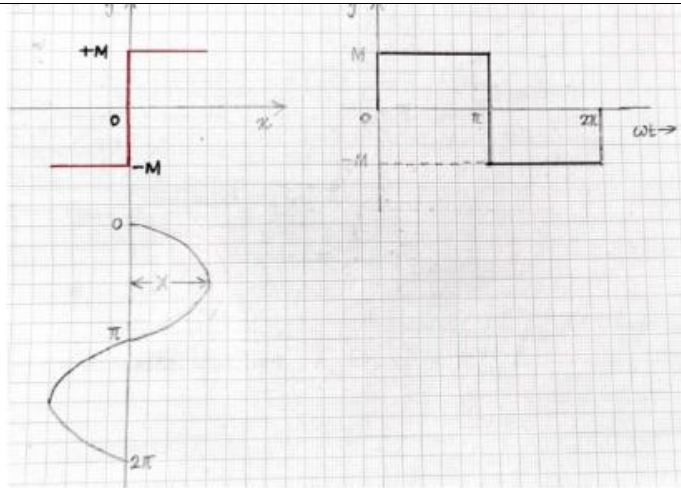
a) Derive the describing function of ideal relay non-linearity given below.



b) Determine the frequency and nature of the limit cycle for the unity feed back system given below.



Ans



Input signal: $x = X \sin \omega t$

Output signal:

$$y(\omega t) = \begin{cases} +M & ; 0 < \omega t < \pi \\ -M & ; \pi < \omega t < 2\pi \end{cases}$$

Describing function :

$$A_1 = \frac{4M}{\pi} \quad B_1 = 0 \quad \phi_1 = 0^\circ$$

Derivation of constants with necessary steps

$$K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \quad Y_1 = \sqrt{(A_1^2 + B_1^2)} \quad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right)$$

$$K_N(X, \omega) = \frac{4M}{\pi X} \angle 0^\circ$$

b

Frequency and nature of the limit cycle:

Plot of $G(j\omega)$

$$\frac{-1}{K_N(X, \omega)} \text{ locus}$$

$$G(j\omega) = \frac{10}{s(s+1)(s+2)} = \frac{5}{j\omega(1+j\omega)(1+j0.5\omega)}$$

$$|G(j\omega)| = \frac{5}{\omega \sqrt{(1+\omega^2)} \sqrt{(1+0.25\omega^2)}} \quad \angle G(j\omega):$$

$$\frac{-1}{K_N(X, \omega)} = \frac{\pi X}{4} \angle -180^\circ$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}(\omega) - \tan^{-1}(0.5\omega)$$

ω rad/s		
$ G(j\omega) $		
$\angle G(j\omega)$		-180°

$G(j\omega)$ locus and $\frac{-1}{K_N(X, \omega)}$ locus intersect at -180° axis

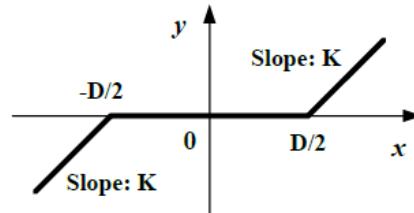
$$-90 - \tan^{-1}(\omega) - \tan^{-1}(0.5\omega) = -180^\circ$$

$$\therefore \omega = \sqrt{2} \text{ rad / s}$$

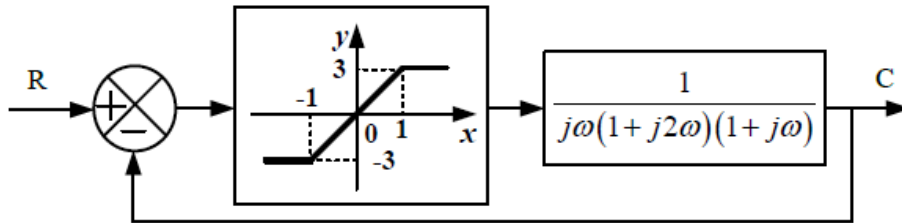
Nature of the limit cycle: stable

Q18

a) Derive the describing function of dead-zone non-linearity given below.



b) Determine the frequency and nature of the limit cycle for the unity feed back system given below.

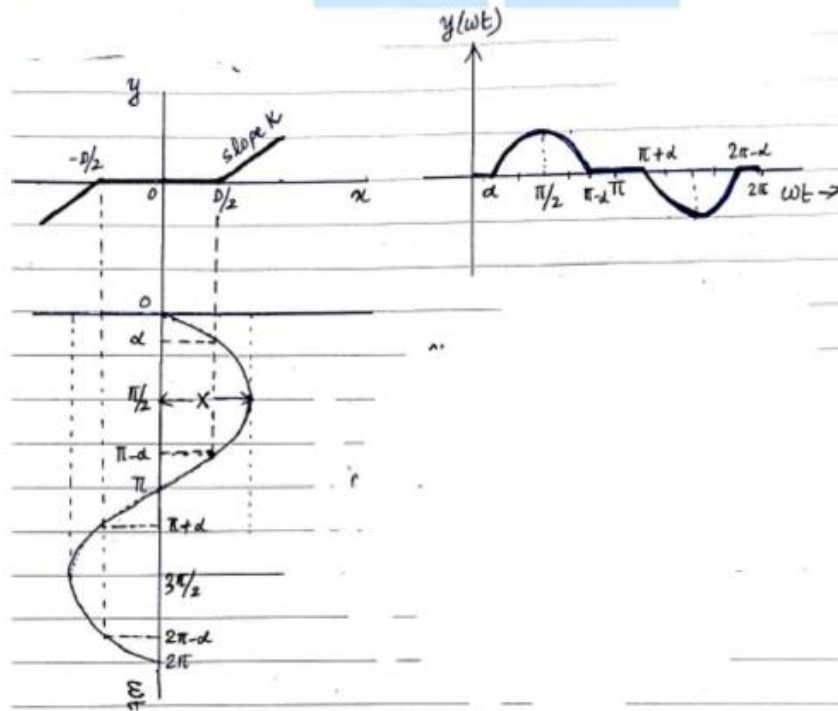


Ans

a

Describing function of dead-zone non-linearity:

Waveforms :



Input signal: $x = X \cdot \sin \omega t$

Output signal:

$$y(\omega t) = \begin{cases} 0 & ; 0 < \omega t < \alpha \\ K \left(x - \frac{D}{2} \right) & ; \alpha < \omega t < (\pi - \alpha) \\ 0 & ; (\pi - \alpha) < \omega t < \pi \end{cases}$$

Describing function :

$$K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \quad Y_1 = \sqrt{A_1^2 + B_1^2} \quad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right)$$

$$A_1 = \frac{4K}{\pi} \left[\frac{X}{2} \left(\frac{\pi}{2} - \alpha + \frac{\sin 2\alpha}{2} \right) - \frac{D}{2} \cos \alpha \right] = KX \left[1 - \frac{2}{\pi} (\alpha + \sin \alpha \cdot \cos \alpha) \right]$$

$$B_1 = 0 \quad \phi_1 = 0^\circ$$

b

Frequency and nature of the limit cycle:

Plot of $G(j\omega)$

$$\frac{-1}{K_N(X, \omega)} \text{ locus}$$

$$G(j\omega) = \frac{1}{j\omega(1+j2\omega)(1+j\omega)}$$

$$|G(j\omega)| = \frac{1}{\omega\sqrt{(1+4\omega^2)}\sqrt{(1+\omega^2)}} \quad \angle G(j\omega) = -90^\circ$$

$$\frac{-1}{K_N(X, \omega)} = \begin{cases} \frac{\pi}{2K\pi[\beta + \sin\beta \cdot \cos\beta]} \angle -180^\circ & ; X > 1 \\ \frac{1}{K} \angle -180^\circ & ; X < 1 \end{cases}$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}(2\omega) - \tan^{-1}(\omega)$$

$\omega \text{ rad/s}$		0.707
$ G(j\omega) $		
$\angle G(j\omega)$		-180°

$G(j\omega)$ locus and $\frac{-1}{K_N(X, \omega)}$ locus intersect at -180° axis

$$-90 - \tan^{-1}(2\omega) - \tan^{-1}(\omega) = -180^\circ$$

$$\therefore \omega = 0.707 \text{ rad/s}$$

Nature of the limit cycle: stable

Module 5

Q19	a) What do you mean by phase trajectory? Explain how to draw the phase trajectory using isocline method. b) Compute the Lyapunov function, $V(X)$ for which the system given below is asymptotically stable. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$
Ans	a) A phase trajectory, also known as a phase plot or phase portrait, is a graphical representation of the behavior of a dynamical system in state space. It shows how the state variables of the system evolve over time, and how they depend on each other. To draw a phase trajectory using the isocline method, follow these steps: Step 1: Determine the system's equilibrium points. An equilibrium point is a state where the system remains unchanged over time. It is the point where the derivative of the state variables is zero. To find the equilibrium points, set the derivative equations of the state variables to zero and solve for the state variables. Step 2: Draw the isoclines. An isocline is a curve in state space along which the derivative of one of the state variables is constant. To draw the isoclines, set the derivative of one state variable to a constant value and plot the resulting curve in state space. Step 3: Determine the direction of the arrows. The direction of the arrows on the phase trajectory indicates the direction in which the state variables change over time. The direction of the arrows can be determined by examining the sign of the derivative of the state variables along the isoclines. Step 4: Sketch the phase trajectory. Sketch the phase trajectory by connecting the equilibrium points and the isoclines. The phase trajectory shows the qualitative behavior of the system in state space, including the stability and the types of oscillations or transient behavior. The isocline method is a useful technique for drawing phase trajectories of nonlinear systems with two state variables. However, it may not be applicable for more complex systems or systems with more than two state variables. In such cases, numerical methods such as simulation or bifurcation analysis may be used. b)

	Lyapunov function: $-Q = A^T \cdot P + P \cdot A$ $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix}^T \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix}$ On solving, $P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = \begin{bmatrix} 23/60 & -7/60 \\ -7/60 & 11/60 \end{bmatrix}$ Positive definiteness of P for stability. $ p_{11} > 0$ $\begin{vmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{vmatrix} = \frac{17}{300} > 0$ Lyapunov function: $V(X) = X^T \cdot P \cdot X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{23}{60} x_1^2 - \frac{14}{60} x_1 x_2 + \frac{11}{60} x_2^2$
Q20	a) Construct the phase trajectory of the second order system given below using isocline method. $\dot{x}_2 + (2\xi\omega_n)x_2 + (\omega_n^2)x_1 = 0$ $X(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1.5 \\ 0 \end{bmatrix} \quad \xi = 0.15 \quad \omega_n = 1 \text{ rad/s}$ where x_1 and x_2 are the state variables. b) Explain Lyapunov function? Using Lyapunov function, $V(X)$, state the condition/s for the system to be (i) stable (ii) asymptotically stable in the large and (iii) unstable. Consider the system described by, $\dot{x}_1 = x_2$ $\dot{x}_2 = -x_1 - x_2^3$ Check the stability of the system for the Lyapunov function, $V(X) = x_1^2 + x_2^2$.

Ans	a To construct the phase trajectory of the second-order system using the isocline method, we first need to find the equilibrium points and draw the isoclines. Equilibrium points: To find the equilibrium points, we set the derivative equations of the state variables to zero and solve for the state variables. For the given system, the state variables are x_1 and x_2. Taking their derivatives with respect to time gives: $\frac{dx_1}{dt} = \xi_1 x_1 + \xi_2 x_2 = 0$ Setting these equations to zero gives: $x_1 = 0, x_2 = 0$ Therefore, the equilibrium point is (0,0). Isoclines: An isocline is a curve in state space along which the derivative of one of the state variables is constant. For the given system, we can find the isoclines by setting the derivative of each state variable to a constant value and solving for the other variable. Isocline for x_1: $\frac{dx_1}{dt} = \xi_1 x_1 + \xi_2 x_2 = k$ Setting the derivative of x_1 to a constant value, say k, we get: $\xi_1 x_1 + \xi_2 x_2 = k$ This is a quadratic equation in x_2, and we can solve for x_2 using the quadratic formula: $x_2 = \frac{-\xi_1 x_1 \pm \sqrt{\xi_1^2 x_1^2 - 4\xi_2(k - \xi_1 x_1)}}{2\xi_2}$ So the isocline for x_1 is a pair of straight lines given by: $x_2 = \frac{-\xi_1 x_1 \pm \sqrt{\xi_1^2 x_1^2 - 4\xi_2(k - \xi_1 x_1)}}{2\xi_2}$ Isocline for x_2: $\frac{dx_2}{dt} = \xi_3 x_1 + \xi_4 x_2 = k$ Setting the derivative of x_2 to a constant value, say k, we get: $\xi_3 x_1 + \xi_4 x_2 = k$ This is also a quadratic equation in x_1, and we can solve for x_1 using the quadratic formula: $x_1 = \frac{-\xi_3 x_2 \pm \sqrt{\xi_3^2 x_2^2 - 4\xi_4(k - \xi_3 x_2)}}{2\xi_4}$ So the isocline for x_2 is a pair of straight lines given by: $x_1 = \frac{-\xi_3 x_2 \pm \sqrt{\xi_3^2 x_2^2 - 4\xi_4(k - \xi_3 x_2)}}{2\xi_4}$ Direction of arrows: The direction of arrows on the phase trajectory indicates the direction in which the state variables change over time. We can determine the direction of arrows by examining the sign of the derivative of the state variables along the isoclines. For the given system, the derivative of x_1 is negative along the isoclines for x_1, and the derivative of x_2 is negative along the isocline
	b Let X be the state vector of a dynamic system and $V(X)$ be a scalar function defined over the state space of the system. $V(X)$ is a Lyapunov function if the following conditions are satisfied: $V(X)$ is positive definite, i.e., $V(X) > 0$ for all $X \neq 0$. $V(X)$ is continuous in the state space. The time derivative of $V(X)$ along the trajectories of the system is negative semi-definite, i.e., $dV(X)/dt \leq 0$ for all $X \neq 0$. Using a Lyapunov function, we can determine whether a system is stable, asymptotically stable, or unstable. (i) Stable: A system is said to be stable if for any initial state X_0, the trajectories of the system converge to a bounded region in the state space. A Lyapunov function $V(X)$ can be used to show that a system is stable if it is positive definite and $dV(X)/dt \leq 0$ for all $X \neq 0$.

	(ii) Asymptotically stable in the large: A system is said to be asymptotically stable if for any initial state X_0, the trajectories of the system converge to a unique equilibrium point in the state space, and this equilibrium point is globally attractive. A Lyapunov function $V(X)$ can be used to show that a system is asymptotically stable in the large if it is positive definite, $dV(X)/dt \leq 0$ for all $X \neq 0$, and $V(X) \rightarrow 0$ as $\ X\ \rightarrow \infty$. (iii) Unstable: A system is said to be unstable if for some initial states X_0, the trajectories of the system diverge to infinity in the state space. A Lyapunov function $V(X)$ can be used to show that a system is unstable if there exists at least one X_0 such that $V(X_0) > 0$ and $dV(X)/dt > 0$ for all $X \neq 0$. Now, let's consider the given system: We want to check the stability of the system for the Lyapunov function $V(X) = x_1^2$. Taking the time derivative of $V(X)$, we have: $dV(X)/dt = 2x_1 dx_1/dt$ Substituting the dynamics of the system, we get: $dV(X)/dt = -2x_1^2 - 4x_1 x_2$ Since $dV(X)/dt$ is negative definite, the system is stable. Note that we cannot determine whether the system is asymptotically stable or not using this Lyapunov function
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Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree (S, FE) Examination May 2024 (2019 Scheme)

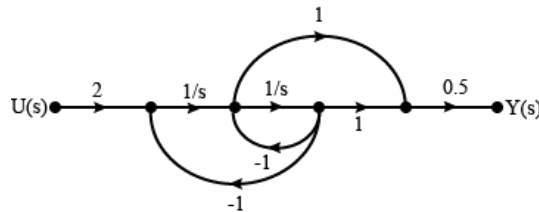
Course Code: EET401**Course Name: ADVANCED CONTROL SYSTEMS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

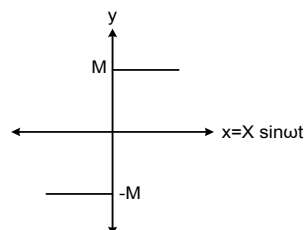
- 1 Obtain the system matrix of the system with the transfer function (3)

$$\frac{Y(s)}{U(s)} = \frac{1}{S^3 + 4S^2 + 5S + 3}$$

- 2 Derive the state model of the system described by the signal flow graph shown below (3)



- 3 Explain Cayley-Hamilton theorem with an example (3)
- 4 Derive the transfer function of an LTI system whose state model is given as (3)
 $\dot{x} = Ax + Bu$ and $y = Cx + Du$.
- 5 Discuss the conditions for controllability of the systems using the Gilberts method. (3)
- 6 Give the block diagram of system described by the state model, $\dot{x} = Ax + Bu$ (3)
 and $y = Cx$ employing a state feedback controller with control law $u = -kx$
 where k is the state feedback controller gain matrix.
- 7 State the assumptions made in the describing function method of analysis. (3)
- 8 Derive the describing function for the non-linearity shown below (3)



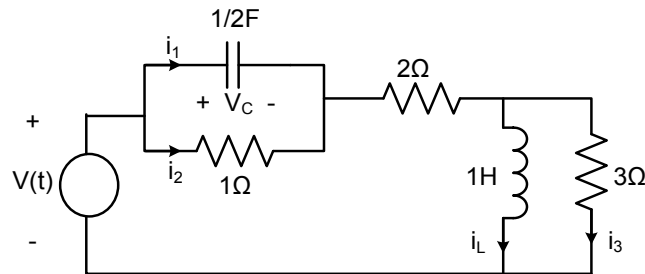
- 9 List the limitations of Lyapunov method of stability analysis. (3)
- 10 Explain the conditions to check the positive and negative definiteness of a quadratic function given as $(x) = X^T A X$, where A is an $n \times n$ real, symmetric matrix. (3)

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

- 11 a) Derive the state model of the electrical circuit shown below (9)

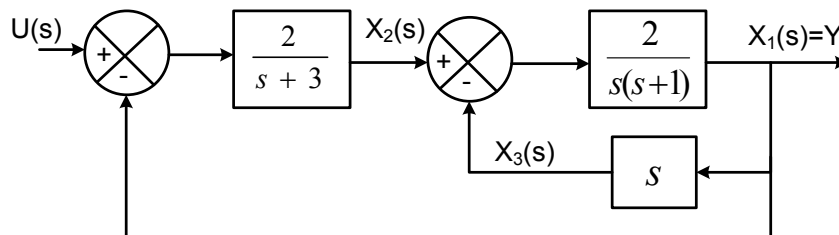


- b) Determine the canonical state model of the system with the transfer function (5)

$$\frac{Y(s)}{U(s)} = \frac{10(S + 4)}{S(S + 1)(S + 3)}$$

OR

- 12 a) Derive the state model of the system shown in the block diagram given below (7)



- b) Using the similarity transformation, diagonalize the matrix $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ (7)

Module II

- 13 a) Obtain the unit step response of the system describe by the state model given below for the initial condition, $x(0) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ (7)

$$\dot{x} = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 0]x$$

- b) Compute the unit step response of the discrete time system described by the state model given below with initial condition as $x(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ (7)

$$\begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix} x(k) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = [-1 \quad 1]x(k)$$

OR

- 14 a) Determine the transfer function of the system describe by the state model given below (8)

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 2 \quad 0]x$$

- b) Derive the equation for the solution of homogeneous state equation. What is state transition matrix? (6)

Module III

- 15 a) Obtain the restriction on a_1 and a_2 to ensure the controllability of the system described by the state equation given below (7)

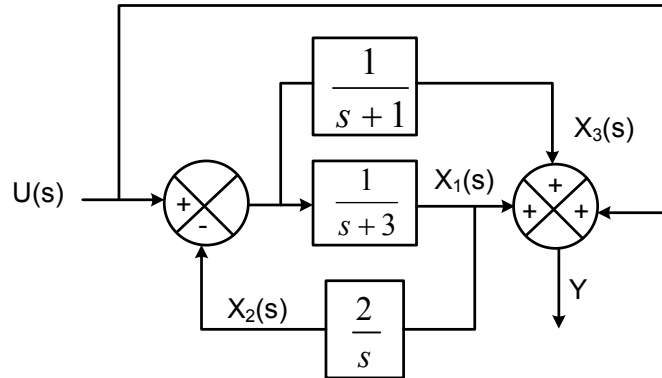
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- b) A regulator has double integrator plant $\frac{Y(s)}{U(s)} = \frac{1}{s^2}$. Compute K such that the control law $U = -KX$ gives closed loop characteristics roots with $\omega_n = 1 \text{ rad/sec}$ and $\tau = \frac{\sqrt{2}}{2}$ (7)

OR

- 16 a) Test the controllability and observability of the system shown in the block diagram given below (8)



- b) Design a full order state observer for the system described in the state model given below (6)

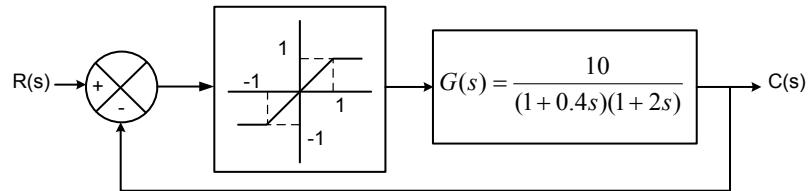
$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 20.6 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 0] x$$

The desired eigenvalues of the observer matrix are $\mu_1 = -8$ and $\mu_2 = -8$

Module IV

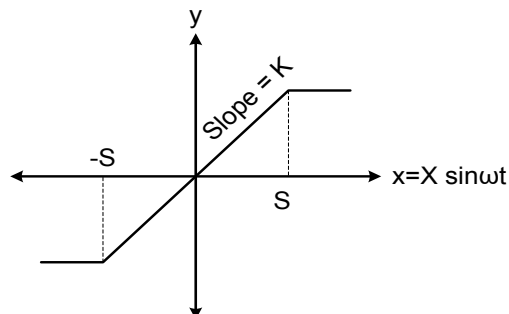
- 17 a) Determine the stability of the system shown below (10)



- b) How the non-linearities are classified? Give examples (4)

OR

- 18 a) Derive the describing function of the nonlinearity shown below (9)



- b) How the stability of the systems are ascertained using describing function method? What are stable and unstable limit cycles? (5)

Module V

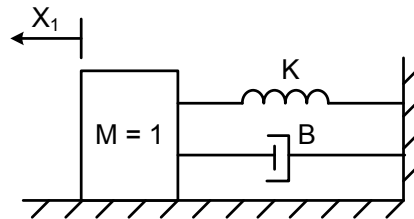
- 19 a) Discuss the asymptotic stability and instability definitions based on the phase plane analysis of non linear systems. (4)
- b) Determine the singular point of the system described by the state equation given below and construct a partial phase trajectory using isocline method. Choose the initial condition, $x(0) = \begin{bmatrix} 1.5 \\ 0 \end{bmatrix}$ (10)

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 - 0.3x_2$$

OR

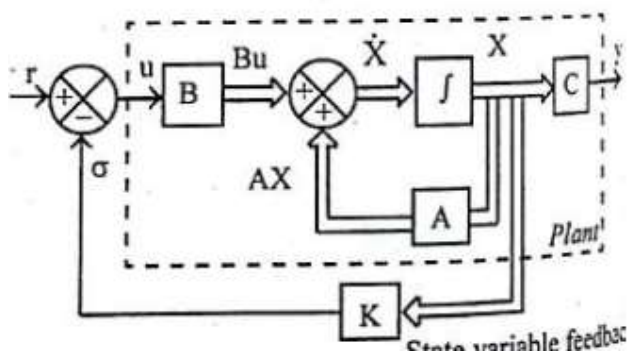
- 20 a) A spring-mass-damper-system shown below. Choose an energy function including the potential energy of the spring and the kinetic energy of the moving mass. Analyse the system stability using Lyapunov method based on concept of stored energy. (8)



- b) Determine the condition for the asymptotic stability of the system described by the state equation given below using Lyapunov method (6)

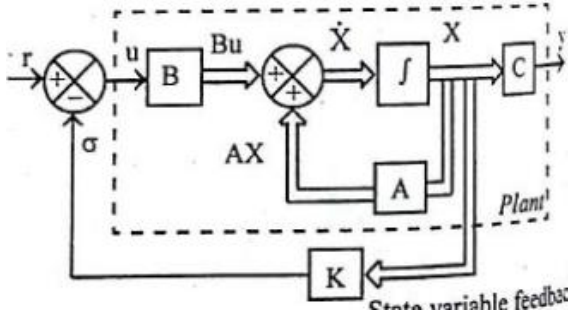
$$\dot{x}_1 = -x_1 + 2x_1^2 x_2$$

$$\dot{x}_2 = -x_2$$

DRAFT Scheme of Valuation/Answer Key (Scheme of evaluation (marks in brackets) and answers of problems/key)		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY SEVENTH SEMESTER B. TECH(S,FE) DEGREE EXAMINATION, MAY 2024 (2019 Scheme)		
Course Code: EET401		
Course Name: ADVANCED CONTROL SYSTEMS		
Max. Marks: 100		Duration: 3 Hours
PART A		
	Answer all questions, each carries 3 marks.	Marks
1	$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -5 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad [A] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -5 & -4 \end{bmatrix}$	(3)
2	$\dot{x} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$ $y = [0 \quad 0.5]x$	(3)
3	Theorem-1.5, Example-1.5	(3)
4	$TF = C[SI - A]^{-1} B + D$	(3)
5	Canonical model-1 Conditions-2	(3)
6		(3)
7	Output contains only the fundamental component, non-linearity have symmetry, and the linear part of the systems have low pass filter characteristics(1+1+1)	(3)
8	Derivation- 2 Waveforms- 1	(3)
9	Difficult to come up with suitable Lyapunov function , conditional stability (1.5+1.5)	(3)
10	Sylvester's criteria for positive and negative definiteness- 1.5 marks each	(3)
PART B		

<i>Answer one full question from each module, each carries 14 marks.</i>			
		Module I	
11	a)	$\dot{x} = \begin{bmatrix} -12 & 6 \\ 5 & 5 \end{bmatrix} x + \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} u$	(9)
	b)	$\dot{x} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$, $y = \begin{bmatrix} 40 \\ 3 & -15 & 5 \end{bmatrix} x$	(5)
		OR	
12	a)	$\dot{x} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} u$, $y = [1 \ 0 \ 0]x$	(7)
	b)	diagonalised matrix $A = \begin{bmatrix} -1 & 0 \\ 0 & 5 \end{bmatrix}$	(7)
		Module II	
13	a)	STM = $\begin{bmatrix} e^{-t} & e^{-t}-e^{-2t} \\ 0 & e^{-2t} \end{bmatrix}$ (3marks) $y(t) = -\frac{1}{2} + \frac{1}{2}e^{-2t} - 2e^{-2t}$ (4marks)	(7)
	b)	STM = $\begin{bmatrix} -3^k & 0 \\ 0 & -2^k \end{bmatrix}$ (3marks) $y(k) = \frac{1}{4}(-3)^k - \frac{1}{3}(-2)^k + \frac{1}{12}u(k)$ (4marks)	(7)
		OR	
14	a)	$TF = \frac{2s+1}{s^3+3s^2+2s}$	(8)
	b)	Derivation of the solution of homogeneous state equation-4 marks What is state transition matrix?- 2 marks	(6)
		Module III	
15	a)	$Q_c = \begin{bmatrix} 0 & 0 & a_1 a_2 \\ 0 & a_2 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ $ Q_c = -a_1 a_2^2$ For Q_c to be nonsingular $ Q_c \neq 0$ $\therefore a_1 \neq 0$ and $a_2 \neq 0$	(7)
	b)	State model-2 Controllability check-1, $u = -[1 \ \sqrt{2}]x - 4$ marks	(7)
		OR	
16	a)	$\dot{x} = \begin{bmatrix} -3 & -1 & 0 \\ 2 & 0 & 0 \\ 0 & -1 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} u$, $y = [1 \ 0 \ 1]x + u$	(8)

		Controllable and not observable State model (2 marks), controllability(3marks), Observability(3marks)	
	b)	$K_e = \begin{bmatrix} 16 \\ 84.6 \end{bmatrix}$	(6)
		Module IV	
17	a)	Polar plot- 4 marks $-\frac{1}{K_N}$ plot- 4 Marks System is stable – 2 marks	(10)
	b)	2 types- 2, examples- 2	(4)
		OR	
18	a)	Schematics- 2, derivation steps -5 , final equation- 2 marks	(9)
	b)	Stability conditions-3 marks , stable and unstable limit cycle- 2 marks	(5)
		Module V	
19	a)	Stability and instability definitions- 2marks each	(4)
	b)	Singular points-2 marks, plot- 8 marks (Minimum 4 points)	(10)
		OR	
20	a)	Select lypanuv function – 2 marks state model- 2 marks stability- 4 marks $V(x) = \frac{1}{2}x_2^2 + \frac{1}{2}Kx_1^2$ and $\dot{V}(x) = -Bx_2^2$ Stable for $B > 0$	(8)
	b)	Steps-2 Stability – 4 marsk Stable for $(1 - 2x_1x_2) > 0$	(6)

Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
B.Tech Degree 7th semester (S,FE) Exam April 2025 (2019 Scheme)			
Course Code: EET401			
Course Name: ADVANCED CONTROL SYSTEMS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions, each carries 3 marks.	Marks
1		Minimum set of variables, variables associated with energy storage elements, examples – 3 marks	(3)
2		$\dot{x} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$, $y = \begin{bmatrix} 40 \\ 3 & -15 & 5 \end{bmatrix} x$	(3)
3		1+1+1	(3)
4		Steps-2 final-1	(3)
5		Modified state model-1 block diagram-2	(3)
			
6		Controllability-1.5, observability-1.5	(3)
7		Stability and instability Conditions (1.5+1.5)	(3)
8		2 types- 2, examples- 1	(3)
9		1+1+1	(3)
10		Indefinite ..Steps-1 answer -2	(3)
PART B			
Answer one full question from each module, each carries 14 marks.			
		Module I	
11	a)	$\dot{x} = \begin{bmatrix} -10 & -2 \\ 1 & -1 \end{bmatrix} x + \begin{bmatrix} 6 \\ 0 \end{bmatrix} u$	(9)

	b)	- characteristic equation of this matrix is given by $\lambda^2 - 3\lambda + 2$ Eigenvalues of matrix A are $\lambda_1 = -1, \lambda_2 = -2$ The corresponding eigenvectors of the eigenvalues λ_1 and λ_2 are $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ respectively - The modal matrix formed by these eigenvectors is given by $\mathbf{P} = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}$ - the diagonal matrix $\mathbf{\Lambda}$ is obtained by the transformation $\mathbf{P}^{-1}\mathbf{AP}$ $\mathbf{\Lambda} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$	(5)
		OR	
12	a)	$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{K_1}{M_1} & \frac{K_1}{M_1} & -\frac{B_1}{M_1} & \frac{B_1}{M_1} \\ \frac{K_1}{M_2} & -\frac{(K_1 + K_2)}{M_2} & \frac{B_1}{M_2} & -\frac{(B_1 + B_2)}{M_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{M_1} \\ 0 \end{bmatrix} [u]$ $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$	(7)
	b)	So, $\dot{x}_1 = -x_1 - u$ $\dot{x}_2 = -(x_2 + x_1) = -(x_2 - x_1 - u)$ $\dot{x}_2 = x_1 - x_2 + u$ $y = \dot{x}_2$ $y = x_1 - x_2 + u$ $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$ $\dot{x} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} x + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$	(7)
		Module II	
13	a)	STM = $\begin{bmatrix} 3e^{-2t} - 2e^{-3t} & e^{-2t} - e^{-3t} \\ -6e^{-2t} + 6e^{-3t} & -2e^{-2t} + 3e^{-3t} \end{bmatrix}$ (3marks) $y(t) = \frac{1}{6} - \frac{1}{2}e^{-2t} + \frac{1}{3}e^{-3t}$ (4marks)	(8)

	b)	$f(\lambda) = e^{\lambda t} = \alpha_0 + \alpha_1 \lambda$ $\therefore f(-1) = e^{-t} = \alpha_0 - \alpha_1$ $\frac{d}{d\lambda} f(\lambda)_{\lambda=-1} = \frac{d}{d\lambda} e^{\lambda t}_{\lambda=-1} = \frac{d}{d\lambda} (\alpha_0 + \alpha_1 \lambda)$ $\Rightarrow te^{-t} = \alpha_1$ - Substituting the value of α_1 in equation(4) we get $\alpha_0 = (1+t)e^{-t}$ <u>Step-3:</u> - The required result is $f(\mathbf{A}) = e^{\mathbf{A}t} = \alpha_0 \mathbf{I} + \alpha_1 \mathbf{A}$ $= \begin{bmatrix} (1+t)e^{-t} & te^{-t} \\ -te^{-t} & (1-t)e^{-t} \end{bmatrix}$	(6)
		OR	
14	a)	$TF = \frac{1}{s^3 + s^2 + 1}$	(8)
	b)	Theorem -3 marks, Example -3 marks	(6)
		Module III	
15	a)	$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$ $\mathbf{Q}_c = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B}]$ $\therefore \mathbf{Q}_c = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & a_3 \\ 1 & a_3 & a_2 + a_3^2 \end{bmatrix}$ $\therefore \mathbf{Q}_c \neq 0$ $\therefore$ The system is controllable.	(7)
	b)	Controllability check-2, $u = -[29.6 \quad 3.6]x - 5$ marks	(7)
		OR	
16	a)	$\dot{x} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} u, \quad y = [1 \quad 0 \quad 0]x$ Controllable and observable	(8)

		State model (2 marks), controllability(3marks), Observability(3marks)	
	b)	$K_e = \begin{bmatrix} 120.6 \\ 20 \end{bmatrix}$	(6)
		Module IV	
17	a)	Polar plot- 4 marks $-\frac{1}{K_N}$ plot- 3 Marks Frequency- 1.41 rad/sec Amplitude – 2.14 (1.5 each)	(10)
	b)	Output contains only the fundamental component, non-linearity have symmetry, and the linear part of the systems have low pass filter characteristics	(4)
		OR	
18	a)	Schematics- 2, derivation steps -4 , final equation- 2 marks	(8)
	b)	2+2+2	(6)
		Module V	
19	a)	Stable and unstable limit cycles- 2marks each	(4)
	b)	$K > 0$ Steps -5 final answer - 5	(10)
		OR	
20	a)	Singular points (0,0) and (2,0) State model-2 marks singular points -2 each $\begin{aligned} x_1 &= x \\ x_2 &= \dot{x}, \end{aligned}$ the corresponding first-order system is given by $\frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -5x_1 - 4x_2 + 2.5x_1^2 \end{bmatrix}.$ • Identify the singular points We then compute the singular points for such a system: they are defined by the equations: $x_2 = 0$, $-5x_1 + 2.5x_1^2 = 0$, thus leading to the two singular points (0, 0) and (2, 0).	(6)
	b)	Asymptotically stable $\dot{V}(x) = -2x_1^4 - 2x_2^2$	(8)
